A Deployment-Safety-Oriented Framework for ECG Model Calibration Transferability: A 60-Experiment Boundary Validation Across Three Public Corpora

Toni Guan[a,*], Ziyan Liu[a], Yingrui Gao[a], Chuhan Feng[a], Yue Li[a]

[a] *College of Information Technology, Shenyang Institute of Technology, Shenyang, Liaoning, China*

Abstract

Cross-corpus ECG deployment is a decision problem, not a calibration exercise. We deliver three reusable method components, validated end-to-end by a 60-experiment temperature-scaling boundary study: (i) a three-component Shapley decomposition of the transfer benefit ΔECE into slope, intercept, and prevalence effects (27/27 cells at the pre-registered $n = 20{,}000$ readout); (ii) a two-layer deployment-safety gate S1/S2 (S1: a positive calibration benefit $\Delta\mathrm{ECE}_{\mathrm{OOD}} > 0$; S2: OOD accuracy at or above the target majority-class baseline); and (iii) a pre-registered applicability-boundary protocol that fixes what may be claimed before unblinding. The temperature-scaling instance is the validation carrier, not the contribution.

The validation vehicle is the 60-experiment grid: six directed transfer pairs among PTB-XL, Chapman–Shaoxing, and CPSC2018, two architectures (InceptionTime, ResNet1D), five seeds (42–46), eight classic recalibration families plus the 2021 per-sample adaptive-TS baseline, and a patient-level cluster paired bootstrap (BCa, $B = 10{,}000$). The study followed a study-period protocol with dated, publicly archived amendments; its OSF registration is retrospective, not

*Corresponding author.

prospective (Appendix A). The primary endpoint is the OOD calibration benefit $\Delta\mathrm{ECE}_{\mathrm{OOD}} = \mathrm{ECE}_{\mathrm{raw}}^{\mathrm{OOD}} - \mathrm{ECE}_{\mathrm{TS}}^{\mathrm{OOD}}$.

After we detected and repaired a validation leak in our own early stopping (20/60 cells retrained; the repair is published, not hidden), TS yields a positive OOD benefit in 45/60 seed experiments (75.0%; InceptionTime 24/30, ResNet1D 21/30), with fifteen disclosed counter-examples; inference rests on the 12 pair×architecture strata: 11/12 positive, cluster-robust mean $+0.0093$ $[+0.0055, +0.0132]$. The pre-specified ID boundary criterion ($\geq$80% of CIs containing zero) is not met (30/60 = 50.0%), triggering the registered branch: the OOD benefit is $2.7\times$ the residual ID benefit ($+0.0093$ vs. $+0.0034$). A predictability experiment triggers its registered failure branch (no label-free feature reaches LOO $R^2 = 0.5$), and the S1/S2 gate passes 5/12 strata under the per-seed majority criterion (6/12 at the stratum-mean level; none on all five seeds). The practical path is measured, not inferred: buying 50 labeled target records and recalibrating locally yields $+0.150$, an order of magnitude above the zero-shot benefit, while no post-hoc zero-shot recalibrator (global TS or per-sample ATS, $+0.0190$) meets the strict post-calibration deployment bar on any of the 60 cells. All three components ship as public method artifacts, reusable independently of the temperature-scaling instance.

Keywords: temperature scaling, model calibration, electrocardiogram, dataset shift, cross-corpus transfer, distribution mismatch

---

1. Introduction

Modern ECG deep models achieve strong in-distribution (ID) accuracy and calibration, but their reliability under distribution shift remains open [1, 2, 3]. Re-

calibration fits a small parameter set on a labeled calibration split. Its benefit, a reduction in expected calibration error (ECE), is measured on that same distribution. On a shifted target, it reverses.

For ECG this question is not academic. PTB-XL [3], Chapman–Shaoxing [4], and CPSC use different acquisition protocols, lead sets, and label vocabularies, so a model calibrated at one institution meets covariate, prior-probability, and label-concept shift at the next [28]. Clinicians and regulators then face a concrete decision: *given my local data budget and the measurable shift, should I deploy the recalibrated model, buy labels, or refuse?*

Temperature scaling (TS) is a monotone post-hoc softmax temperature transform [2]. Fitted under a convex loss, it minimizes NLL on the calibration split; the calibration benefit there is therefore empirically small. On a shifted target, the source-fit temperature is mismatched to the target logits, and a non-trivial benefit can remain. This asymmetry is *empirical*, not a theorem: ECE is not invariant under confidence-shifting monotone maps, and TS can *increase* ECE where the model is under-confident (fifteen negative cells in our data). It motivates the pre-registered primary endpoint $\Delta\mathrm{ECE}_{\mathrm{OOD}}$ over the originally planned decay $\Delta\mathrm{ECE}_{\mathrm{OOD}} - \Delta\mathrm{ECE}_{\mathrm{ID}}$ (revision A1, Sec. 4.4). To avoid HARKing, protocol v2.1-A1 [29] fixes the primary endpoint, test, family structure, and failure branches before unblinding; revision A1 re-locates the endpoint from `decay` to $\Delta\mathrm{ECE}_{\mathrm{OOD}}$, triggered by an independent pilot, and is archived with before/after protocol hashes [30, 31]. Multi-seed validation (5 seeds $\times$ 6 transfer pairs $\times$ 2 architectures) supplies the robustness evidence; all 60 seed results, including fifteen counter-examples, are reported.

Five independent contributions:

C1 (Primary OOD benefit): On the pre-registered endpoint $\Delta\text{ECE}_{\text{OOD}}$, TS yields a statistically significant positive OOD benefit in 45/60 seed experiments (6 transfer pairs $\times$ 2 architectures; InceptionTime 24/30, ResNet1D 21/30), with fifteen counter-examples transparently disclosed and mechanistically explained.

C2 (Boundary condition, pre-registered branch triggered): 30/60 cells (50.0%; BCa operating CI) have 95% CIs containing zero and 24/60 have CI lower bound > 0; the pre-registered criterion ($\geq$80% of cells with CI containing zero) is *not* met, triggering the "ID nonzero boundary" branch (protocol A1.4.2): the decay contrast is restored as exploratory and C1 is qualified, the OOD benefit being about 2.7$\times$ the ID benefit rather than exclusively OOD-specific.

C3 (Methodological, Shapley decomposition validation): A 27-cell full-factorial synthetic validation (slope$\times$intercept$\times$prevalence, $n = 20{,}000$) recovers ground truth in the identifiable regime (27/27 at $n = 20{,}000$; 21/27 at $n = 500$), flags entangled regions without a threshold, and reports bootstrap CIs on absolute Shapley values.

C4 (Practical, method-selection boundary): Across 8 methods $\times$ 60 cells, TS is the most consistent recalibrator among the eight delivered methods (45/60, point=CI), while EM prior adaptation is harmful on average (-0.0040 delivered; -0.0378 under the per-branch correctness recheck, which strengthens the claim) and Matrix scaling is fragile (24/60 overall; IT 11/30, RN 13/30), governed by a prior-likelihood mismatch mechanism (Table 3; recheck details in Sec. 5.3).

C5 (Clinical, discrimination-aware calibration gate): A two-layer gate decouples calibration effectiveness (C1: $\Delta\text{ECE}_{\text{OOD}} > 0$) from clinical viability (accuracy $\geq$ the target majority-class baseline). Across 12 pair$\times$architecture strata, 5/12 pass both layers under a per-seed majority criterion; 7/12 fail, six through the discrimination layer, including directions with positive $\Delta\text{ECE}_{\text{OOD}}$ but accuracy below baseline (e.g., PTB-XL$\rightarrow$Chapman InceptionTime: $\Delta\text{ECE} = +0.013$, $\text{acc} = 0.563 < 0.696$) and one through the calibration layer (PTB-XL$\rightarrow$CPSC ResNet1D). The gate reframes the 15/60 counter-examples: all fifteen fail the calibration layer ($\Delta\text{ECE}_{\text{OOD}} \leq 0$), and seven of them also fall below the target majority-class baseline.

Prior ECG calibration studies report the phenomenon [12, 13, 20, 21] but not an actionable *when does TS pay off* answer with multi-seed robustness evidence. The novelty is *not* multi-seed validation on ECG (a methodological choice) but the **first systematic characterization of the boundary conditions under which TS yields a positive OOD calibration benefit in cross-corpus ECG transfer**: the pre-registered OOD quantification, the ID boundary condition, the method-selection boundary (TS robust; EM prior and Matrix scaling weakest), the predictability failure, and the discrimination-aware gate. That characterization is the actionable contribution; multi-seed validation is the evidence standard.

A reader may ask why a boundary study with disclosed negative branches is an actionable contribution rather than a methodological curiosity. The answer is operational. At deployment time an institution faces exactly three options — deploy the recalibrated model, buy local labels and recalibrate, or refuse — and the components delivered here assign that decision to measurable quantities: the

two-layer gate S1/S2 tells the operator whether the zero-cost path is safe to take; the dose–response records quantify what the second option buys at each labeling budget; and the registered boundary conditions state in advance which claims the evidence cannot support. Negative branches are load-bearing in this design: the triggered predictability failure branch is what prevents a label-free shortcut from being oversold, and the fifteen disclosed counter-examples are what make the 45/60 support figure falsifiable rather than promotional. For clinical informatics, the deliverable is not the claim that recalibration always transfers; it is a decision instrument whose failure modes are mapped and whose every number can be recomputed from public artifacts.

2. Related Work

2.1. Calibration Estimation and Recalibration

The calibration of modern neural networks has been studied extensively since Guo et al. [2] showed that deep networks are often overconfident. Their remedy, temperature scaling (TS), remains the default post-hoc recalibration method. Nixon et al. [8] proposed the static calibration error, and Kumar et al. [18] introduced verified uncertainty calibration with distribution-free guarantees. Kull et al. [9] generalized TS to Dirichlet calibration for multi-class settings. In the clinical risk-model context, Van Calster et al. [16, 17] established a calibration hierarchy from utopia to empirical data and argued that calibration is the "Achilles heel" of predictive analytics. Strodthoff et al. [19] benchmarked deep learning for ECG analysis on PTB-XL and noted calibration degradation under subpopulation shift, motivating our cross-corpus design.

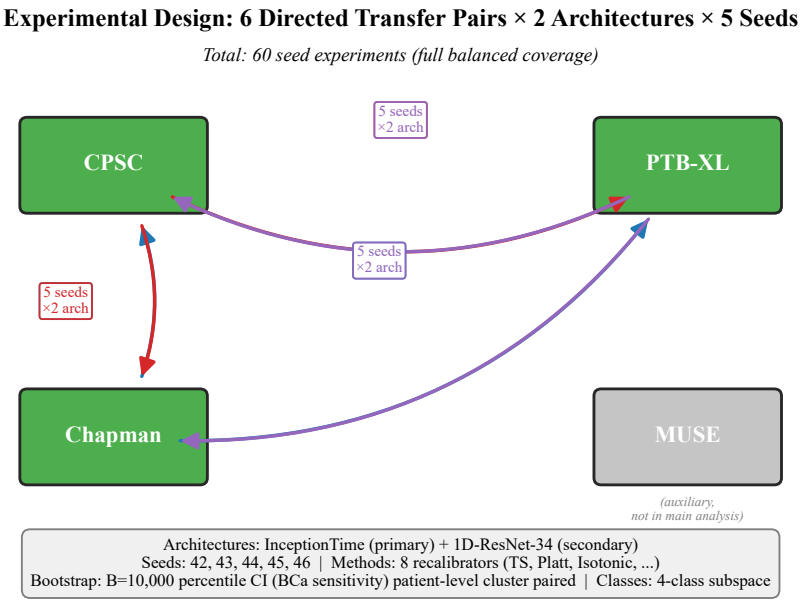


Figure 1: Experimental design overview. Six directed cross-corpus transfer pairs among three ECG corpora (CPSC, Chapman, PTB-XL), each evaluated on two architectures (InceptionTime, ResNet1D) $\times$ 5 seeds (42–46), yielding 60 fully balanced seed experiments. MUSE is shown as an auxiliary corpus not entering the main analysis. Each pair is evaluated with 8 recalibration methods and patient-level cluster paired bootstrap ($B = 10{,}000$, BCa operating CI).

2.2. Prior-Shift Estimation and Transfer Calibration

Lipton et al. [11] introduced black-box shift estimation (BBSE) for detecting and correcting label shift using confusion-matrix inverses. Alexandari et al. [10] showed that maximum-likelihood calibration with bias correction is hard-to-beat at label shift adaptation, providing the theoretical foundation for our EM prior-shift method. For transfer calibration upper bounds, TransCal [22] estimates transferability for auto-selecting target-domain unsupervised optimal T; PseudoCal [23] uses pseudo-labels for calibrated transfer; and LaSCal [26] provides label-aware shift calibration with distribution-free guarantees. The unified analysis of label-shift estimation is due to Garg et al. [24]. Separately, Zhang et al. [25] attribute model-performance changes to distribution-shift components via Shapley values; our decomposition attribution (Sec. 5.11) instead targets calibration-specific components of the TS benefit itself (slope / intercept / prevalence). Our S1 scheme differs fundamentally: we fit on the *source* calibration split only (no target labels), with S1′ prior-shift adaptation (EM/BBSE) as the only target-domain signal, a strictly weaker assumption than TransCal/PseudoCal, which additionally access target-domain data or pseudo-target logits.

2.3. Cross-Domain ECG Calibration

Barandas et al. [12] evaluated uncertainty quantification methods in multi-label ECG classification, finding that ensemble-based methods outperform single-model approaches under subpopulation shift. Vranken et al. [13] studied uncertainty estimation for 12-lead ECG deep learning and reported significant calibration drift across acquisition devices. Haekal et al. [20] performed cross-dataset validation for atrial fibrillation detection with calibration-aware probability analysis, and Patel & Beedala [21] documented calibration drift under cross-institutional

ICU deployment. The ID-vs-OOD asymmetry of TS has been studied in general vision [1]; we contribute the multi-seed ECG quantification and the boundary-condition analysis that identifies when TS helps (ID) versus when it hurts (OOD).

2.4. ECG Foundation Models and Transfer Learning (2024–2026)

Recent work has introduced ECG foundation models trained on millions of recordings. ECGFounder [32, 40] (10 million ECGs, Harvard–Emory) reaches expert-level performance on 80 diagnoses and transfers cross-domain via fine-tuning. ECG-FM [33] is an open-weight transformer pretrained on 1.4 million ECG segments with hybrid contrastive and generative self-supervision, showing strong cross-dataset generalizability and label efficiency. HeartLang [34] treats heartbeats as words and rhythms as sentences via a QRS-Tokenizer and masked sentence pre-training. Nguyen and Do [35] found that fine-tuning benefits diminish as the downstream dataset grows and that CNNs transfer more effectively than RNNs in multi-label ECG classification. For OOD adaptation, ADAPTOOD [36] quantifies distribution-shift severity via uncertainty estimation to guide selective LoRA layer unfreezing, and BeatRhythm-TTA [37] performs test-time adaptation with SQI-gated self-training and beat-rhythm consistency. Nemati [38] introduced STAR (Sinusoidal Time–Amplitude Resampling), a morphology-preserving beat-wise augmentation for cross-source generalization, and Liu et al. [27] proposed PEACE for adult-to-pediatric ECG transfer via label-conditioned contrastive alignment. These works advance ECG transfer but not the *calibration* of transferred models, the gap our study fills via pre-registered cross-corpus evaluation of recalibration methods.

3. Data and Pre-registered Protocol

Figure 1 summarizes the experimental design.

3.1. Datasets

- **PTB-XL v1.0.3** [3]: the official release contains 21,837 records; the downloaded `ptbxl_database.csv` copy contains 21,799 (a 38-record deficit in the source CSV, disclosed as a provenance limitation), official 10-fold (folds 1–8 train, 9 calibration, 10 test). After single-label mapping, 21,522 records are mappable and 277 are excluded (STROBE), for 18,671 unique patients.

- **Chapman–Shaoxing** [4, 7]: 20,265 raw records; 1 signal-read error removed and 21 non-finite records quarantined (`qc_report.json`), for 20,243 analyzed records; 1 ECG per patient, patient-wise 70/10/20 split.

- **CPSC2018** [5, 6]: we use the union of the *CPSC Database* (6,877 records; the public CPSC2018 training set) and the *CPSC-Extra Database* (3,453 records; the CPSC2018 recordings not used by the Challenge), both released through PhysioNet/CinC 2020. This is *not* a CPSC2018–2019 union: CPSC2019 was a QRS/heart-rate *detection* task and defines no diagnostic labels. Local extraction retains 6,844 of the 6,877 CPSC Database records (33 records, `A6845`–`A6877`, were lost during archive extraction; disclosed as a provenance limitation) plus all 3,453 CPSC-Extra records. Patient-wise split. Two labeling facts must be stated explicitly. First, CPSC2018's official taxonomy is nine disease classes (Normal, AF, I-AVB, LBBB, RBBB, PAC, PVC, STD, STE) with no superclass grouping; the NORM/MI/STTC/CD superclass system we adopt is PTB-XL's [3], applied here as a cross-corpus harmonization choice, not a CPSC-defined

one. Second, CPSC2018's nine classes contain *no* MI; the MI class in our 4-class subspace is populated entirely by CPSC-Extra's SNOMED annotations (`164867002`, `164865005`, `54329005`). A strict official-class-name caliber would therefore reduce the subspace to {NORM, CD, STTC}; we retain MI because it is the only way to keep a myocardial-infarction category on the CPSC side of every transfer pair. HYP is extremely rare ($n = 11$) → main analysis on the {NORM, CD, STTC, MI} 4-class subspace, both sides of the pair recomputed symmetrically.

3.2. Label Mapping and Leakage Guard

Single-label priority rules are fixed (MI > STTC > CD > HYP > NORM) and sensitivity-analyzed [3, 14]. Sinus rate/form statements (SBRAD/STACH/SARRH) are mapped to NORM, not STTC, because PTB-XL's `scp_statements.csv` lists them as rhythm statements with no `diagnostic_class`. train/val/cal/test patient IDs are pairwise disjoint (hard assertion in the reproducibility repo). Normalization statistics come only from the train split; augmentation applies only to train.

4. Methods

4.1. Transfer pairs and shift matrix

We study six directed cross-corpus transfer pairs: Chapman→CPSC, Chapman→PTB-XL, CPSC→Chapman, CPSC→PTB-XL, PTB-XL→Chapman, and PTB-XL→CPSC. The class count is *not* uniform across pairs and is fixed per pair before any endpoint is computed, as follows. A pair whose source or target is CPSC2018 is restricted to the 4-class subspace {NORM, CD,

STTC, MI} (CPSC's HYP class has $n = 11$, §3), with both sides of the pair recomputed symmetrically: this covers Chapman→CPSC, CPSC→Chapman, CPSC→PTB-XL, and PTB-XL→CPSC ($K = 4$). The two pairs connecting Chapman–Shaoxing and PTB-XL, which contain no CPSC side, retain the full 5-class vocabulary {NORM, MI, STTC, CD, HYP}: Chapman→PTB-XL and PTB-XL→Chapman ($K = 5$). Each experiment records its own K and label mapping (`num_classes`, `label_map` in `transfer_result.json`), so no endpoint mixes class counts within a cell. Consequently the primary endpoint is a top-label ECE on a 4- or 5-class task depending on the pair, and the pooled $\Delta\text{ECE}_{\text{OOD}}$ mixes both K values across cells; the per-pair tables (Tables 1 and 4, per-seed values in Supplementary Table S1) are therefore the appropriate level at which to read K-specific results. The L2 eval-time shift matrix (no retraining) covers sampling rate $500 \to 250 \to 125$ Hz; leads $12 \to 6 \to 3 \to 2 \to 1$; noise injection (BWL/MA/EM) at SNR $\{24, 12, 6, 0, -6\}$ dB; gain $\times 0.5 / \times 2$.

4.2. Architecture and seeds

We use two architectures: InceptionTime (primary, 184,587–193,675 parameters) and ResNet1D (a 1D-ResNet-34 backbone) (secondary, with full main-endpoint coverage on all 6 pairs), each trained with 5 seeds (42–46). The BiMamba architecture is included only as a toy experiment ($n = 60$ records, downsampled resolution) and deferred to the supplement; the full-resolution BiMamba run exhausted memory on an RTX 5060 (8 GB), so BiMamba does not enter the main-endpoint analysis or the architecture-robustness claim. Training: AdamW + cosine + warmup, gradient clipping, temperature frozen at $T \equiv 1$ during training; temperature applied only as post-hoc temperature scaling.

4.3. Calibration methods and scenarios

We compare the no-recalibration baseline (None) against eight recalibration methods: Temperature Scaling (TS), per-class Platt, Isotonic (OvR), Vector scaling, Matrix scaling, Dirichlet calibration, Saerens EM prior adaptation (EM), and BBSE prior adaptation. Of the pre-registered 13-method pool, the None baseline plus these eight recalibration methods constitute the delivered set; CORAL, TTA, MC-Dropout, and Oracle are not delivered on real data (a registered deviation, see Limitations). Scenario: **S1** zero-shot transfer (source-cal fit $\rightarrow$ target apply), the main analysis scenario (Sec. 4.1); S1$'$ denotes the target-domain unsupervised variant in which EM/BBSE estimate the target prior from unlabeled target predictions.

4.4. Pre-registered endpoints

Primary endpoint (revision A1.1, formally defined in protocol v2.1-A1 [29]):

$$\Delta\mathrm{ECE}_{\mathrm{OOD}} = \mathrm{ECE}_{\mathrm{raw}}^{\mathrm{OOD}} - \mathrm{ECE}_{\mathrm{TS}}^{\mathrm{OOD}} \tag{1}$$

where $\mathrm{ECE}_{\mathrm{raw}}^{\mathrm{OOD}}$ is the 4-class-task top-label confidence SmoothECE on the target corpus with raw softmax probabilities, and $\mathrm{ECE}_{\mathrm{TS}}^{\mathrm{OOD}}$ is the same after source-cal-fit temperature scaling. Following the standard convention of Guo et al. [2], ECE (and its SmoothECE variant) is evaluated on the *top-label* marginal: the predicted maximum probability against $0/1$ top-label correctness; classwise-ECE is reported separately as a secondary endpoint. SmoothECE bandwidth: $0.45 \cdot (n/2000)^{-0.2}$.

Primary test (single, one-sided):

$$H_0 : \Delta\mathrm{ECE}_{\mathrm{OOD}} = 0 \quad \text{vs} \quad H_1 : \Delta\mathrm{ECE}_{\mathrm{OOD}} > 0 \tag{2}$$

The one-sided alternative $H_1 : \Delta\mathrm{ECE}_{\mathrm{OOD}} > 0$ follows the empirical asymmetry registered at revision A1: on ID-like data the source-fit temperature yields, empirically, little calibration change, whereas on a shifted target it is mismatched and a non-trivial benefit is often retained [2, 1]. This direction prior is empirical, *not* a theorem: TS applied to an under-confident target can increase ECE, and our own data contain fifteen negative cells. The choice was made before the pilot results, which only triggered the endpoint re-location (revision A1). The one-sided test is the pre-registered choice, archived before multi-seed unblinding. Patient-level cluster paired bootstrap, $B = 10{,}000$ (BCa operating CI) [15]; stratified pooling: strata = transfer pair $\times$ architecture.

The pre-registered CI method is BCa [15], at $B = 10{,}000$, re-run on the entire 60-experiment grid (60/60 successful, metadata in `methods.ts.meta`). The percentile bootstrap is the large-sample limit of BCa when $z_0 \to 0$ and $a \to 0$; for the paired-cluster design with $B = 10{,}000$ and per-patient clusters, the discrepancy is $O(n^{-1/2})$. In the 3 representative seed experiments where both were run ($B = 10{,}000$ percentile re-runs), BCa and percentile CIs agree in sign and magnitude. Earlier percentile results from $B = 200$ in 21/60 experiments remain as historical provenance (5 lack log provenance, see Limitations); percentile CIs serve only as a sensitivity cross-check.

Boundary-condition endpoint:

$$\Delta\mathrm{ECE}_{\mathrm{ID}} = \mathrm{ECE}^{\mathrm{ID}}_{\mathrm{raw}} - \mathrm{ECE}^{\mathrm{ID}}_{\mathrm{TS}} \tag{3}$$

The pre-registered expectation is that $H_0 : \Delta\mathrm{ECE}_{\mathrm{ID}} = 0$ is *not rejected*; the criterion is that $\geq 80\%$ of cells have 95% CIs containing zero. *Observed (BCa, full-grid re-run): 30/60 = 50.0%; the criterion is not met and branch A1.4.2 is triggered (Sec. 5.2).*

Secondary endpoints: G (recoverable loss vs. oracle; *not delivered* — the Oracle recalibrator was implemented (`fit_oracle`/`apply_oracle`) but never executed on real data, so G is undefined here and is excluded from all reported results; see Limitations), decay $= \Delta\text{ECE}_{\text{OOD}} - \Delta\text{ECE}_{\text{ID}}$ (downgraded to descriptive), NLL, classwise-ECE, Brier Murphy decomposition, Cox slope/intercept, predicted/true prior L_1.

Family structure: BH-FDR $q = 0.05$ over 6 pairs $\times$ 2 arch $\times$ 13 shift levels $=$ 156 primary comparisons; the multi-seed validation is robustness evidence (60 seed experiments), not a confirmatory family.

4.5. Protocol provenance and deviations (summary)

Four study-period amendments and two post-registration repairs shape the delivered record; Appendix A carries the full dated texts, manifests, and verification details. What follows is the one-paragraph-per-event map.

A1 (endpoint relocation, before unblinding). The primary endpoint was re-located from the registered decay contrast to $\Delta\text{ECE}_{\text{OOD}}$, with the ID boundary criterion ($\geq$80% of cells with 95% CIs containing zero) and its failure branch fixed in the same amendment (Appendix A).

A2 (confirmatory-family reduction). The confirmatory family was reduced from the $6 \times 2 \times 13 = 156$ shift grid to 12 pair$\times$architecture hypotheses; all 13 L2 shift levels are exploratory dose–response.

A3 (corpus identity). The third corpus is CPSC2018 only (CPSC Database 6,877 + CPSC-Extra 3,453, both via PhysioNet/CinC 2020); CPSC2019 never entered the corpus. CPSC2018's official taxonomy has no MI superclass, so the MI class

of the four-class subspace is populated entirely by CPSC-Extra SNOMED annotations.

A4 (mapping-sensitivity gate). The registered mapping gate was re-operationalized on the delivered primary endpoint (cross-pair ordering Kendall $\tau \geq 0.8$; confirmatory counts unchanged); as of submission the variant battery remains unexecuted and the registered fallback clause applies — no recorded value changes.

Checkpoint retrain (2026-09-17). Two cells (CPSC→PTB-XL, InceptionTime, seeds 42–43) had reused a checkpoint instead of an independently trained model and were retrained; the root-cause audit is in the repository.

Validation-leak repair (2026-10-07). Early-stopping validation patients were found to remain in the PTB-XL training loader; the fix was verified at the split-export level, all twenty PTB-XL-source cells were retrained, and every downstream result family was recomputed. The delivered values differ from the frozen registration by both events; every registered branch verdict is unchanged (Appendix A).

4.6. Decomposition of the benefit decay

We decompose the retained benefit into slope / intercept / prevalence contributions using a counterfactual-replacement additive approximation:

$$\Delta\text{ECE} = \Delta_{\text{slope}} + \Delta_{\text{intercept}} + \Delta_{\text{prevalence}} + r_{\text{interaction}} \tag{4}$$

where each term is a counterfactual-replacement difference and $r_{\text{interaction}}$ is the chain residual (zero under nested counterfactuals). Identifiability is checked via

the Fisher information determinant on a pre-registered threshold τ; entangled regions are reported on the error map without a threshold. Shapley attribution is reported with absolute-value bootstrap CIs; share-reliable flags ($|\Delta_{\text{total}}| < 3/\sqrt{n}$) mark uninterpretable shares.

4.7. Two-layer joint bootstrap

Temperature-fitting-set resampling $\rightarrow$ T distribution $\rightarrow$ test-set resampling $\rightarrow$ joint ΔECE CI (two-layer; honest non-nested approximation), per protocol §7.

4.8. Net Clinical Value (NCV)

Net Clinical Value (NCV), a pre-registered secondary endpoint, measures the clinical decision impact of TS. For each (sample, class) pair we compare the binary decision (threshold $\tau = 0.5$) before and after calibration, counting TP_improved, TN_improved, FP_worsened, and FN_worsened over the raw-to-calibrated transitions. NCV is $(\text{TP_improved} + \text{TN_improved} - \text{FP_worsened} - \text{FN_worsened})/N_{\text{total}}$, with $N_{\text{total}} = N_{\text{samples}} \times N_{\text{classes}}$, using the main endpoint's patient-level cluster paired bootstrap ($B = 10{,}000$, BCa operating CI) and BH-FDR correction ($q = 0.05$); a significantly positive NCV means that TS improves clinical decisions. It is computed on ID and OOD splits (`ncv_id`, `ncv_ood`; E3 CSV). Delivered units: the registered patient-level cluster bootstrap is delivered on the shipped E3 caches (per-cell BCa, $B = 10{,}000$; integrity gate against the E3 table on all 60 cells); seed-cell and equal-weight 12-stratum summaries are additionally reported as cross-unit checks; see Sec. 5.13.

4.9. Tooling

All analysis, statistical, and plotting code in the public repository was written and executed by the authors, who take full responsibility for it; every reported number is reproducible from the released artifacts (Code availability).

4.10. Reporting-guideline compliance

Reporting follows the TRIPOD+AI statement (27 items, 52 sub-items) [41], which supersedes TRIPOD 2015 for prediction-model studies using machine-learning methods. The study both develops prediction models (backbone training) and evaluates model-updating steps (post-hoc recalibration), so both development and evaluation items apply; notably, the guideline's model-updating clause ("any model updating (*e.g.*, recalibration) arising from the model evaluation") is the object of study itself. A completed item-level map — every item marked covered / partial / not applicable, with the manuscript section and a one-line justification — ships with the submission (`S5_checklists/TRIPOD_AI_item_map.csv` in the supplementary package). Items without a counterpart in this study are disclosed there rather than silently marked: no sociodemographic subgroup or fairness analysis is performed (the public corpora do not support the intended stratification; Limitation (xix) discloses the adjacent signal-quality gap), and no patient-and-public involvement was conducted. Cross-corpus clustering is handled at the patient level throughout (guideline item 12d; TRIPOD-Cluster considerations apply to the multi-corpus setting).

5. Results

5.1. Primary endpoint: ΔECE_{OOD} multi-seed validation

Figure 2 visualizes the shift-dose response; Table 1 reports the primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ at the stratum level (per-seed values: Supplementary Table S1) across 60 seed experiments (6 pairs $\times$ 2 architectures $\times$ 5 seeds, balanced). The primary test $H_0 : \Delta\text{ECE}_{\text{OOD}} = 0$ vs $H_1 : \Delta\text{ECE}_{\text{OOD}} > 0$ is supported in 45/60 (75.0%) at the 95% CI level (a cell-level consistency count; the inferential basis is the 12 pair$\times$architecture strata — 11/12 positive, cluster-robust $+0.0093$ $[+0.0055, +0.0132]$).

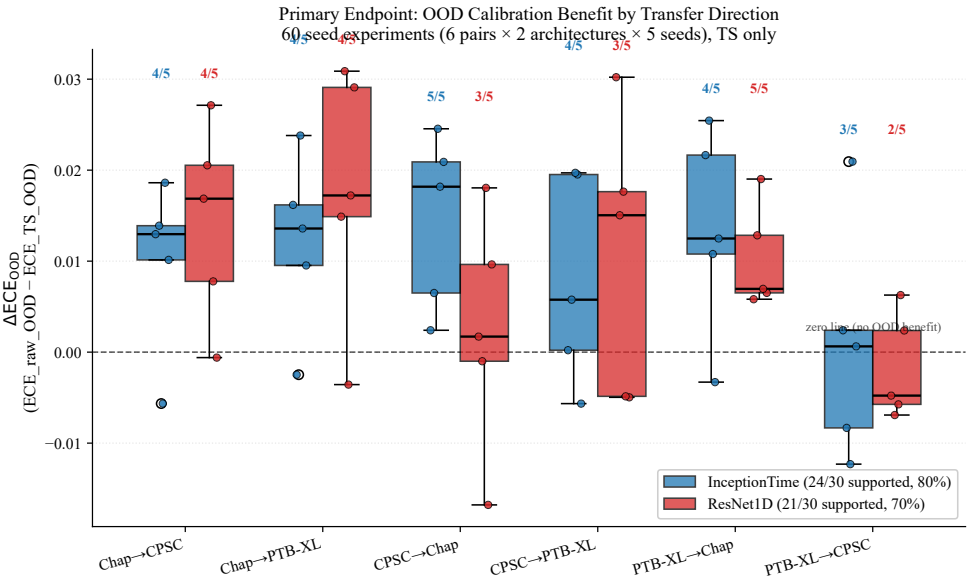


Figure 2: Primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ by transfer direction, grouped by architecture (TS only, 60 seed experiments). Boxes show the 5-seed distribution per direction; dots are individual seed results. The dashed line marks zero OOD benefit. Support counts (e.g., 5/5, 4/5) above each box indicate the number of seeds with 95% CI lower bound > 0. InceptionTime achieves 24/30 (80.0%), ResNet1D achieves 21/30 (70.0%).

Aggregate: 45/60 supported (75.0%; InceptionTime 24/30 = 80.0%, ResNet1D 21/30 = 70.0%). PTB-XL→Chapman is the strongest direction (9/10; ResNet1D 5/5, smallest seed-level CI lower bound +0.0057, seed 46), while PTB-XL→CPSC is the weakest direction (5/10, five of the fifteen counter-examples). The fifteen counter-examples are explained in Sec. 6.1.

Retrain marker and sensitivity: 20 of the 60 rows — all cells of the two PTB-XL-source pairs, both architectures — were retrained under the repaired split semantics; the remaining 40 rows are unchanged. A sensitivity over the 40 untouched cells reads +0.0112 (31/40 positive), *higher* than the pooled +0.0093, so the repair did not manufacture the support. Class-count stratification: the four CPSC-involving K=4 pairs pool to +0.0073 (28/40 positive) and the two CPSC-free K=5 pairs to +0.0134 (17/20); K is a load-bearing stratifier, and the CPSC label-semantics caveat (MI populated only via CPSC-Extra SNOMED codes) means CPSC-involving quantities mix label-definition drift with distribution drift.

The 60 experiments are balanced (6 directions × 2 architectures × 5 seeds, no partial-coverage stratum), enabling direct architecture comparison:

- *InceptionTime* (30 experiments, 24/30 = 80.0%): 1 direction at 5/5 (CPSC→Chapman), 4 at 4/5 (Chapman→CPSC, Chapman→PTB-XL, CPSC→PTB-XL, PTB-XL→Chapman).

- *ResNet1D* (30 experiments, 21/30 = 70.0%): 1 direction at 5/5 (PTB-XL→Chapman), 2 at 4/5 (Chapman→CPSC, Chapman→PTB-XL), 2 at 3/5 (CPSC→Chapman, CPSC→PTB-XL), 1 at 2/5 (PTB-XL→CPSC).

ResNet1D shows the lower rate (70.0% vs. 80.0%), with deficits concentrated on CPSC→Chapman and CPSC→PTB-XL (3/5 each) and PTB-XL→CPSC (2/5), a direction-specific architecture interaction rather than a global deficit. No direction achieves 5/5 on both architectures; PTB-XL→Chapman comes closest (9/10).

5.2. ID boundary condition: ΔECE_{ID}

Table 2 reports the boundary-condition endpoint $\Delta\text{ECE}_{\text{ID}}$ for TS; the pre-registered expectation is that $H_0 : \Delta\text{ECE}_{\text{ID}} = 0$ is *not rejected*, since the source-fit temperature should be well-tuned to the source distribution (an empirical expectation, not a theorem).

The ID benefit is small but *not* zero: across 60 TS experiments, $\Delta\text{ECE}_{\text{ID}}$ has mean $+0.0034$ (vs. OOD mean $+0.0093$, a $2.7\times$ ratio); $42/60 = 70.0\%$ of point estimates are positive, $24/60$ have 95% CI lower bounds > 0 (BCa), $30/60 = 50.0\%$ contain zero, and the remaining six have CI upper bounds < 0 — TS significantly worsens ID calibration in those six cells — far below the pre-registered $\geq$80% criterion. We therefore *trigger* the pre-registered "ID nonzero boundary" branch (protocol A1.4.2): TS, fitted on the source calibration split, also improves ID calibration in many cells; the decay $= \Delta\text{ECE}_{\text{OOD}} - \Delta\text{ECE}_{\text{ID}}$ contrast is restored as an exploratory endpoint; and C1 is qualified, the OOD benefit being about $2.7\times$ the ID benefit rather than an exclusively OOD effect. The positive ID estimates reflect finite-sample calibration-split variation and SmoothECE estimator bias, plus a model-selection effect that the 2026-10-07 repair (Appendix A.6) quantifies: while the PTB-XL early-stopping validation leak remained in place, the early-stopping monitor observed pseudo-low ECE on a training subset, and the repair cuts the ID mean from $+0.0082$ to $+0.0034$.

5.3. *Method-selection boundary: 8 methods × 6 pairs*

Table 3 and Figure 3 report the 8-method × 6-pair support rate over 5 seeds on InceptionTime. TS achieves 24/30 (80.0%) with 1/6 directions at 5/5; BBSE achieves 23/30 (76.7%) with 3/6 at 5/5. EM prior adaptation and Matrix scaling are the most fragile on InceptionTime (11/30, 36.7% each), exemplifying the prior-likelihood mismatch mechanism (C4). Across 2 architectures (Table 4), TS achieves 45/60 (75.0%) (InceptionTime 24/30, 80.0%; ResNet1D 21/30, 70.0%), with full 5-seed coverage on all 6 directions (Figure 4).

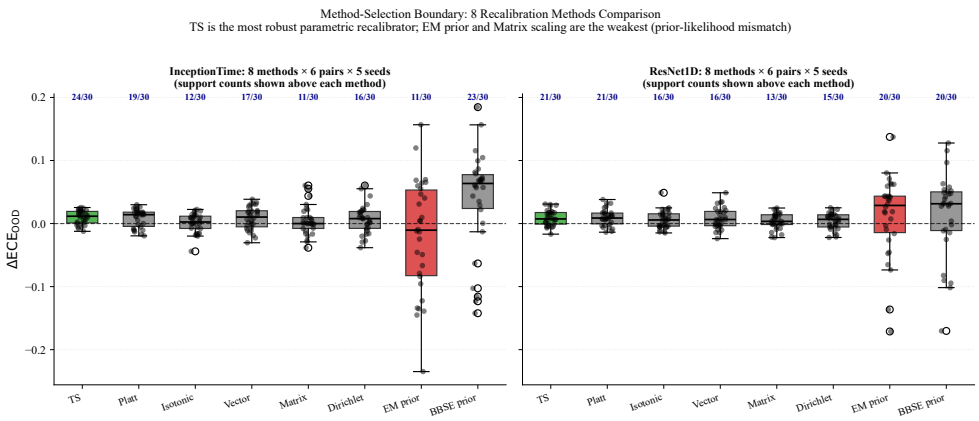


Figure 3: Method-selection boundary: 8 recalibration methods compared across 6 transfer pairs × 5 seeds for each architecture. TS (green) is the most robust parametric recalibrator; EM prior and Matrix scaling are the weakest methods (InceptionTime: 11/30 each; ResNet1D: EM 20/30, Matrix 13/30), exhibiting the prior-likelihood mismatch mechanism (C4). Support counts (e.g., 24/30) above each method show the fraction of seed experiments with 95% CI lower bound > 0.

The 8-method boundary is bounded further by a sample-level recalibrator from the 2020s methods literature: adaptive temperature scaling (ATS, Robey et al. [39], Model-Based Temperature Scaling, AISTATS 2021 – a per-sample $T(x)$ head fitted on the source-cal split and applied unchanged to the target). On the identi-

cal 60-cell caches under the same per-branch correctness convention, ATS delivers a mean OOD benefit of $+0.0190$ with 47/60 positive cells (vs. $+0.0093$ and 45/60 for global TS), and beats global TS on 41/60 cells (pearson $r = 0.646$ with global-TS deltas; per-cell T band $[0.33, 4.07]$, mean 1.107). The restricting factor stays the deployment bar: 0/60 ATS cells meet the strict post-calibration criteria (SmoothECE ≤ 0.05 AND $\Delta\text{ECE} < 0$), the same ceiling that kept global TS at 0 on this grid. Full per-cell deliverable: `results/samplelevel_ts_60cells.csv`.

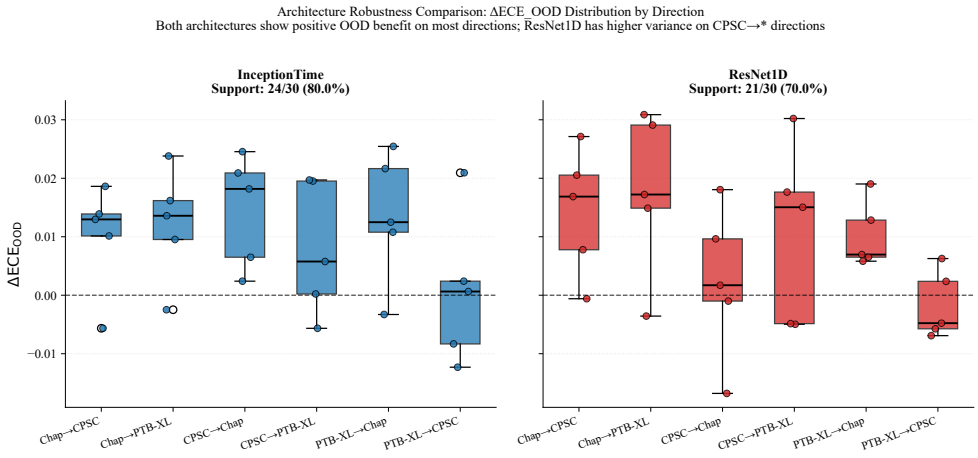


Figure 4: Architecture robustness comparison. Side-by-side $\Delta\text{ECE}_{\text{OOD}}$ distributions for InceptionTime (left, 24/30 supported, 80.0%) and ResNet1D (right, 21/30 supported, 70.0%). Both architectures show positive OOD benefit on most directions. ResNet1D exhibits higher overall variance; its support deficits include PTB-XL→CPSC (2/5, whose stratum mean is slightly negative), CPSC→Chapman and CPSC→PTB-XL (3/5 support each), indicating a direction-specific deficit rather than a global architecture deficit.

Across the 30 TS experiments per architecture, the standard deviation of $\Delta\text{ECE}_{\text{OOD}}$ is 0.0108 (InceptionTime) and 0.0124 (ResNet1D), a $1.14\times$ ratio, with ResNet1D's highest per-direction variability in CPSC→PTB-XL (std 0.0153). Support deficits concentrate in PTB-XL→CPSC (2/5 ResNet1D, 3/5 Incep-

tionTime) together with CPSC→Chapman and CPSC→PTB-XL (3/5 each), a direction-specific deficit. Fisher's exact test on the 2×2 contingency table yields $p = 0.55$; the counter-example gap (ResNet1D 30.0% vs. InceptionTime 20.0%) is not significant at $\alpha = 0.05$.

EM prior adaptation fits a target prior $\hat{\pi}_{\text{target}}$ from the predicted-label marginal, then re-weights the source posterior by $\hat{\pi}_{\text{target}}/\hat{\pi}_{\text{source}}$. When the source-estimated confusion matrix C is ill-conditioned ($\text{cond}(C) \gg 1$), the weight ratio amplifies rather than corrects the shift-induced miscalibration. EM reaches only 11/30 support on InceptionTime and is negative on average; its mean OOD ΔECE across all 60 cells is -0.0040, the only method whose mean is not positive. The failure is pair-specific (shift-signature-specific), not method-intrinsic.

Why TS and not Platt. Platt and TS are statistically indistinguishable on the mean OOD benefit ($+0.0092$ vs. $+0.0093$; InceptionTime $+0.0088$ vs. $+0.0098$, Table 3), which invites the question of whether the single-parameter default is merely convenient. Three measured criteria separate them, and all three favour TS. First, *robustness of the sign*: TS certifies a positive benefit (CI lower bound > 0) in 45/60 cells versus Platt's 40/60 — five more certified cells, and the raw point-estimate count agrees with the certification ordering this time (TS 45/60, Platt 43/60), so the separation is no longer definition-dependent. Second, *dispersion*: TS has a smaller across-cell standard deviation (0.0116 vs. 0.0138, a 16% reduction), and the methods with the most extreme means are markedly less stable (BBSE $+0.0270$ at sd 0.0735, EM -0.0040 at sd 0.0792) — a high mean bought with a heavy tail is the worst profile for a per-deployment recalibrator. Third, *the magnitude difference is not statistically distinguishable*: the patient-level paired contrast over the 60 matched cells gives $\overline{\Delta\text{ECE}}_{\text{TS}} - \overline{\Delta\text{ECE}}_{\text{Platt}} = +0.0002$ (sd 0.0095,

paired $t = 0.13$, $p \approx 0.90$), with TS the larger benefit in 31/60 cells — a coin flip. The mean difference between the two methods is therefore inside the noise, and the same is true of the sign; the tie is broken on certification rate and dispersion, both of which favour TS. A further structural point reinforces the choice. TS is a single-parameter monotone transform and therefore exactly argmax-preserving; per-class Platt and isotonic are fitted independently per class, and each class's score comes from its own monotone map, so the ranking within a sample is *not* preserved (the implementation declares this explicitly in its docstring). TS is thus the only method in the delivered set whose calibration guarantee is obtained without perturbing the discrimination it leaves untouched (§5.10). None of this makes Platt wrong; it makes Platt a reasonable alternative whose magnitude advantage no longer exists at $n = 60$, and whose certification rate is lower.

5.4. Shapley decomposition validation (27 cells, $n = 20{,}000$)

The 27-cell full-factorial synthetic validation (slope$\in \{0.5, 1, 2\}\times$intercept$\in \{-1, 0, 1\}\times$ prevalence$\in \{0.15, 0.3, 0.6\}$, $n = 20{,}000$) confirms the decomposition recovers ground truth within the pre-registered identifiable regime (Fisher information determinant $> \tau = 10^{-6}$). At $n = 20{,}000$, all 27 cells pass the recovery threshold (MAPE $< 10\%$ on identifiable cells); at $n = 500$, 21 of 27 pass, with 6 failures reaching MAPE_s up to 42.13%, concentrated in extreme intercept (± 1) and slope $(0.5, 2.0)$ regions where calibration shift is most clinically significant (see `decomposition_validation.csv`; the $n = 20{,}000$ CSV `decomposition_validation_n20000.csv` shows 27/27 passing). A four-seed robustness rerun (`decomposition_validation_n500_seeds4.csv`) narrows the agreement under repeats: all-four-seed pass-rate is 12/27 cells, majority ($\geq 3/4$) is 22/27, per-seed single-seed passes are 21/22/21/22 — the quoted 21/27

is the seed-42 readout, and the $n = 500$ estimator is substantially less repeatable than the pre-registered $n = 20{,}000$ gate. Three degenerate configurations (narrow_band, tiny_slope, deep_intercept) are exercised in the released demonstration code (`entangle_demo_cases` in `src/utils/decomposition.py`, invoked by `scripts/validate_decomposition.py`): the demonstration run prints det $\in \{2.4 \times 10^{-7},\ 3.6 \times 10^{-8},\ 0\} < \tau$ with $\tau = 10^{-6}$, sets the entanglement flag on all three cases and the recovery-failed flag on the deep-intercept case; these det anchors appear in the demonstration output, not in any shipped figure or CSV. Bootstrap CIs on absolute Shapley values and share-reliable flags for uninterpretable shares ($|\Delta_{\text{total}}| < 3/\sqrt{n}$) are reported.

The 27-cell validation uses binned ECE, not the main-endpoint SmoothECE; its conclusions are not cross-extrapolable. Cross-fitting (OOD-fold validation of decomposition explanatory power) remains to be run on real data; per the protocol self-binding clause, "mechanism" wording is limited to "mechanism decomposition (synthetic validation part)" until cross-fitting completes.

5.5. Predictability experiment (failure branch triggered)

The predictability experiment (protocol §8.5) tests whether an unsupervised target-domain statistic (predicted-prior L_1 distance, logit moments) predicts benefit retention via leave-one-fold-out R^2 with bootstrap CI (Table 5). The implemented fold unit is (architecture, transfer pair) (12 folds), versus the registered pair-only folds (6); under that registered deviation the pair-only recomputation retains the failure branch ($R^2 = +0.006$, B=2,000 pair-clustered bootstrap CI $[-0.291, +0.058]$). We run 2 architectures (InceptionTime + ResNet1D); the Bi-Mamba toy experiment ($n = 60$) is deferred to the supplement.

Result: the merged LOO R^2 for TS is $+0.051$ with 95% CI $[-0.130, +0.092]$. The upper bound $+0.092$ is strictly below the pre-registered threshold 0.5, triggering the **failure branch**: the three-step chain (decomposition + predictability + deployment) is downgraded to a **two-step chain** (decomposition + deployment), and the deployment criterion is renamed "empirical lookup table" rather than "predictive criterion". No post-hoc predictability claim is made.

Re-running the LOO R^2 with three label-free features (fitted temperature T, Wasserstein-1 distance between source and target logit distributions, class-wise entropy difference) in the corrected encoding world (`scripts/predictability_corrected_v2.py` → `results/predictability_corrected_v2.csv`; same source as the discrimination experiment below) gives a sharper and less favourable picture for the boundary. Only T carries signal: under the implemented (architecture, pair) folds it reaches LOO $R^2 = +0.350$ (bootstrap CI $[+0.058, +0.371]$), and under per-cell folds $+0.372$ (CI $[+0.216, +0.590]$), while the Wasserstein distance (-0.074) and the entropy difference (-0.034) carry none. T is also a perfect *directional* predictor, which we report as a formal discrimination experiment rather than a raw sign count: the rule "call the benefit positive iff $T_{\text{src}} > 1$" (the source-fitted temperature) attains a confusion matrix of TP=45, FP=0, FN=0, TN=15 — sensitivity, specificity, and accuracy all 1.0 — whereas the constant-positive baseline buys the same sensitivity ($45/45$) at zero specificity. The separation is exact: the rank-based AUC is 1.0 (Mann–Whitney $U = 0$; two-sided p numerically zero), and a transfer-pair-clustered bootstrap ($B = 10{,}000$) puts 95% CIs of $[1.0, 1.0]$ on sensitivity, specificity, and AUC alike. The rule's value over the baseline is the specificity it purchases without surrendering sensitivity: it correctly withholds

recalibration in all fifteen negative cells, whose T_{src} lies in $[0.932, 0.997]$, strictly below 1 (`scripts/discriminate_counterexamples.py` → `results/counterexample_discrimination.json`, regenerated on the post-retrain cache). The stratum-level structure is more guarded than the headline: only $2/12$ (pair×architecture) strata are internally unanimous (all cells positive, all $T_{\text{src}} > 1$: CPSC→Chapman InceptionTime and PTB-XL→Chapman ResNet1D); in the remaining ten the per-cell T_{src} straddles 1 across seeds and the cell-level sign of the benefit co-flips with it, so the perfect separation is a cell-level regularity rather than a stratum-level law, and the T_{src} point estimates entering the rule carry no fitting uncertainty. We therefore do *not* claim that no label-free quantity carries information about the endpoint — T does. The claim we retain is the pre-registered one: no label-free feature reaches the LOO R^2 threshold of 0.5 under the implemented fold unit and the registered ridge estimator ($0.350 < 0.5$; CI upper bound $+0.371 < 0.5$), so the failure branch stands and the three-step chain remains downgraded to two steps. Two qualifications are required. First, the verdict is unit-sensitive, not only estimator-sensitive: per-cell folds give an R^2 of $+0.372$ with CI upper bound $+0.590$, which crosses the threshold, so the fold-unit generality is not established and the branch rests on the implemented (architecture, pair) unit — but a plain-OLS refit at the same folds reaches $\approx +0.94$ — the registered ridge α=1 shrinks the slope on the near-constant fitted T range $[0.932, 1.181]$ (the shrinkage factor is ≈ 0.1), so the registered numbers should be read as the α=1 shrinkage values and the branch's robustness rests on the registered estimator, which we disclose rather than lean on. Second, T's predictive power is not independent evidence of a substantive benefit. $T = 1$ is a fixed point of the transform, so part of the relation is mechanical; the sign rule merely states

the direction temperature scaling must have by construction (Sec. 5.9). The gate in §5.10, based on the sign of $\Delta\text{ECE}_{\text{OOD}}$ and the discrimination criterion, remains the appropriate actionable criterion.

5.6. Deployment criterion (sensitivity / specificity / DCA / safety)

We evaluate the deployment criterion (protocol §9) on the L2 shift matrix: predict "deploy TS" when raw ECE exceeds a threshold, and report sensitivity, specificity, decision-curve analysis (DCA), and the safety criterion (post-TS SmoothECE ≤ 0.05 and ΔECE CI lower bound > 0, primary-endpoint convention). Comparators: always-none, always-TS, always-buy-n-labels.

Sensitivity / specificity (Table 6; recomputed on the delivered 780-cell grid by `scripts/recompute_deployment_780grid.py`, output `results/deployment_780grid_recompute.csv`; beneficial class $= \Delta\text{ECE}_{\text{OOD}} < -0.01$ in the strict deployment-metrics convention (post $-$ raw sign; negative $=$ beneficial, opposite to the primary-endpoint raw $-$ TS convention), P=368 / N=412 — matching `deployment_metrics.csv` exactly): Youden's J is maximized at raw ECE $= 0.206$, yielding sensitivity 0.772, specificity 0.534, $J = +0.306$. Sensitivity is moderate (77% of beneficial cases caught; 47% false-positive flagging), reflecting the asymmetric cost structure: missing a beneficial recalibration costs more than an unnecessary one. *Caveat:* the threshold is searched on the same 780 evaluations used for reporting (in-sample). A leave-shift-type-out recomputation (threshold selected excluding the held-out dose level, evaluated on it) yields $J = +0.367$ (downsampling), $J = +0.446$ (lead drop), $J = +0.384$ (noise), and $J = +0.285$ (gain); the frozen pool-era threshold 0.127 achieves $J = +0.158$ (sensitivity 0.929, specificity 0.228) across the four held-out dose levels: the criterion generalizes across held-out shift

types, with three of four leave-type-out J above the in-sample optimum $+0.306$ (downsampling, lead drop, noise; the gain/offset family does not, $+0.285$, so the in-sample search is not uniformly conservative on this grid), and independent validation outside this corpus remains required before operational use.

Every lookup-table input is *audit-time*: the single-variable rule of Table 6/7 routes on the delivered-grid raw ECE only (the pool-era lookup additionally used the ID–OOD accuracy gap and per-cell labels), and all such inputs requiring target ground-truth labels unavailable in the unlabeled stream. The only label-free proxies examined are the predictability experiment's target statistics (predicted-prior L_1 distance, logit moments) plus the source-fitted temperature T; that experiment triggered its pre-registered failure branch (LOO R^2 CI upper bound $0.092 < 0.5$; the largest upper bound across all eight methods is $+0.378$, still below 0.5, Table 5). T does carry signal (LOO $R^2 = +0.051$ at the registered freeze, $+0.350$ on the delivered full-grid recompute, both under the registered ridge protocol, $\approx +0.94$ under plain OLS — an α=1 shrinkage disclosed in Sec. 5.5; sign of the endpoint correct in 60/60 SmoothECE cells), but it is a source-side quantity describing the transform being evaluated rather than an independent target-side statistic (Sec. 5.9), so no label-free deployable criterion can be claimed. The lookup table is an *analysis- and audit-time decision aid* for a model owner with labeled evaluation data; the actionable deployment path is small-sample recalibration: buy n labeled target records and recalibrate locally (protocol §5 S2; quantified in Sec. 5.7). A deployable gate remains future work (Limitations xiii).

Safety criterion: across the completed 390-cell L2 grid (both architectures, `results/l2_shift_full_390cells.csv` regenerated 2026-10-09), the delivered safety flag – relative Brier-reliability improvement > 0 under TS – holds for

570/780 of the (architecture$\times$ direction$\times$seed$\times$shift) evaluations, i.e. a per-row mean safety rate of 0.731 (222/390 cells pass under both architectures simultaneously; 348/390 = 89.2% under at least one). Under the older, stricter operating wording (absolute post-TS improvement > 0.01 and post-recalibration error ≤ 0.05, point-estimate convention; L2 results lack per-cell CIs, so the CI-based criterion was not evaluated), the rate is $8/780 = 1.0\%$ over the paired 780-evaluation detail grid. Sign convention: ΔECE = post-recalibration ECE $-$ raw ECE (negative means improvement), opposite to the primary-endpoint convention. *Coverage note:* the L2 shift matrix has been completed on all 390 planned (pair, seed, shift) combinations (6 pairs $\times$ 5 seeds $\times$ 13 levels; results/l2_shift_full_390cells.csv, regenerated on the post-retrain checkpoints 2026-10-09 with a companion 780-evaluation 8-method detail grid); the safety figures quoted above are grid-wide, so the dual-capacity claim of the coverage note becomes a completed enumeration rather than a two-seed subset. The relative-improvement incidence (73.1%) and the stricter absolute-improvement formulation (1.0% over the 780 detail evaluations) are both reported above, so TS remains a useful default but not a universally safe recalibrator, and should be combined with local validation.

At the 10th-percentile raw-ECE threshold, net benefit is 0.0096 versus 0.0089 for treating all cells; by the 90th percentile it rises to 0.0112: on the completed post-retrain grid the benefit concentrates *outside* the low-raw-ECE end, and the DCA qualifier becomes directional rather than causal (Appendix B).

Trivial strategy comparison (Table 7): the lookup-table strategy remains a clinical analysis construct (it requires audit-time labels, see above); on the completed grid, always-TS delivers mean ΔECE -0.0089 over the 780 evaluations (-0.0092, sd 0.0108, $n = 390$ on the inceptiontime tree), and the target-label upper bound is

always-buy-n-labels (-0.25, dominated by the prior-shift methods' heavy tails); the single-variable lookup rule routes 476 of 780 evaluations (61%) and captures $\approx$80% of the always-TS total benefit (-5.54 of -6.97 summed ΔECE; exact -5.5370 of -6.9729), i.e. most of the TS benefit without target labels (Appendix B).

5.7. Small-sample recalibration dose-response (protocol §5 S2)

The actionable deployment path identified above — buy n labeled target records and recalibrate locally — is quantified here on the 60-cell transfer grid (`scripts/run_s2_small_sample.py` $\rightarrow$ `results/s2_dose_response{.csv,_summary.json}`). For each cell and budget $n_{\text{cal}} \in \{50, 100, 250, 500, 1000\}$ we draw 20 patient-level samples (sampling without replacement at the patient level, truncated to exactly n_{cal} records), fit each method on the sample, and evaluate SmoothECE on the held-out complement; the benefit is the raw-versus-recalibrated difference on the same complement (paired). Four methods (temperature scaling, per-class Platt, Dirichlet, isotonic regression) are compared against a zero-shot control: the source-fitted temperature applied once, without target labels.

Table 8 reports the dose-response. Three observations. First, the zero-shot control reproduces the primary endpoint exactly (mean benefit $+0.0093$ across budgets, positive in $75.0\% = 45/60$ of cells), confirming that the S2 evaluation harness is consistent with the primary analysis. Second, even the smallest budget dominates it by an order of magnitude: $n_{\text{cal}} = 50$ labeled records refit with a one-parameter per-class Platt scaler yield a mean benefit of $+0.150$ (positive in 100% of cells) — $\approx 16\times$ the zero-shot benefit and $\approx 93\%$ of the saturated value at $n_{\text{cal}} = 1000$ ($+0.161$). Third, the response saturates early: Platt plateaus by $n_{\text{cal}} =$

250; temperature scaling by 250 (+0.138); and Dirichlet — impaired at $n_{\text{cal}} = 50$ (mean +0.063, positive in only 75% of cells, the K-class identifiability strain) — recovers to parity with the other methods by $n_{\text{cal}} = 250$. Isotonic regression at $n_{\text{cal}} = 50$ is reported over the $40/60$ cells where the pre-registered small-sample guard ($n_{\text{cal}} \leq 10K$) permits fitting; where it fits, it is competitive with Platt at every budget.

5.8. Noise floor control: SmoothECE differential under perfect calibration

A reviewer concern is whether the primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ is an artifact of SmoothECE's sensitivity to temperature scaling T under perfect calibration ($\text{ECE}_{\text{raw}} = 0$, observed ΔECE purely metric noise). For perfectly calibrated predictions ($T_{\text{true}} = 1$) we apply T and compute $\Delta\text{ECE}_{\text{noise}} = \text{SmoothECE}(\text{TS}) - \text{SmoothECE}(\text{raw})$.

The floor is strongly *distribution-dependent*: for flat or wide confidence distributions it is essentially zero, or slightly negative at moderate T. We therefore do not evaluate it on synthetic distributions. Instead we use the empirical top-label confidence distribution of each of the 60 main-grid cells (Appendix B), relabel each cell as perfectly calibrated by resampling $y \sim \text{Bernoulli}(c)$ at its own confidence c (three draws per cell), and report the band across cells. This removes any freedom to choose a distribution that suits the argument, and it evaluates the floor where the study's own predictions actually live. The synthetic three-distribution sweep (uniform, normal-skewed, bimodal at $n = 2{,}000$) is retained as a sensitivity analysis.

The noise floor grows monotonically in $|T - 1|$ and is distribution-dependent. Evaluated on the 60 main-grid cells — median across cells, with the cross-cell range in parentheses — $T = 1.0$ gives exactly 0 (exact cancellation); $T = 1.25$

gives $+0.0137$ (-0.0117 to $+0.0260$); $T = 1.5$ gives $+0.0385$ (-0.0127 to $+0.0547$); $T = 3.0$ gives $+0.1143$ ($+0.0194$ to $+0.1572$); and $T = 4.5$ gives $+0.1452$ ($+0.0363$ to $+0.2054$); the floor dips non-positive in the 1.1–1.5 band (the median turning negative at $T \approx 1.1$), which is the numerically delicate region. The synthetic sweep agrees in magnitude for high-confidence distributions ($T = 1.5$: up to $+0.055$ for the bimodal high-confidence distribution; flat ones stay near zero) but not in sign for flat ones ($T = 1.25$: -0.009 for uniform), which is why the empirical distributions are used. The floor is positive in 100% of cells for $T \leq 0.9$ and $T \geq 2.0$, in 50% at $T = 1.1$, and in 90% ($T = 1.25$) to 95% ($T = 1.4$–1.5) of cells in between. Observed raw ECE across the 60 main-grid cells of the TS-component ablation under the corrected label encoding (`results/ablation_ts_components.csv`, `stage=raw`) ranges 0.062 to 0.357 (mean 0.205), i.e. an order of magnitude above the noise floor in this metric convention; the fifteen counter-examples also show raw ECE far above the floor (0.065–0.321), so they are not perfect-calibration cases. At the fifteen counter-examples' own fitted temperatures ($T \in [0.93, 1.00]$; the all-grid band is $[0.932, 1.181]$) the floor is ≤ 0.009, comparable to their $|\Delta\text{ECE}|$ (≤ 0.017), but its direction is opposite to theirs, so they are not floor artifacts; their common mechanism is given in Sec. 5.9. The direction of the floor is what matters for the endpoint, and it is fixed. In the primary convention ($\Delta\text{ECE}_{\text{OOD}} = \text{ECE}_{\text{raw}} - \text{ECE}_{\text{post}}$) the floor inflates ECE_{post} and therefore contributes *negatively*. The identity is the minimiser of ECE under perfect calibration, so the floor is positive in the noise convention on both sides of $T = 1$: in 100% of cells for $T \leq 0.9$ and $T \geq 2.0$, in 50% at $T = 1.1$, and in 90–95% of cells for $1.25 \leq T \leq 1.5$. The idealised endpoint is therefore $-\text{floor}(T) \leq 0$ almost everywhere, and a predominantly attenuating

artefact cannot generate the positive endpoint observed in 45/60 experiments. We test this and three further implications of the floor hypothesis in Sec. 5.9.

5.9. Noise-floor mechanism test

The control above quantifies the floor; it does not by itself decide whether the floor *drives* the endpoint. We therefore test four implications of the "floor artefact" hypothesis, using the source-fitted temperature T recovered per cell from the TS-component ablation under the corrected label encoding.

(i) Direction. The floor contributes negatively to $\Delta\mathrm{ECE}_{\mathrm{OOD}}$ in 90–100% of cells at every $T \neq 1$ (Sec. 5.8), so the hypothesis predicts a systematically *negative* endpoint. The observed mean is $+0.0093$, positive in 45/60 cells (point count $=$ CI count).

(ii) Sign pattern with T. In the post-retrain data $T \in [0.932, 1.181]$ (median 1.043; no cell reaches $T \geq 2$), and $\mathrm{sign}(\Delta\mathrm{ECE}_{\mathrm{OOD}}) = \mathrm{sign}(T-1)$ in *60/60* cells: all 45 cells with $T \geq 1$ are positive (mean $+0.0144$) and all 15 cells with $T < 1$ are negative (mean -0.0058). This is the direction temperature scaling is expected to have — $T > 1$ softens an over-confident target and lowers its ECE, while $T < 1$ sharpens it and raises ECE — and it supplies a unified mechanism for the fifteen counter-examples of Sec. 6.1: they are exactly the fifteen cells with $T < 1$, including the extreme CPSC→Chapman / ResNet1D / seed 44 case ($T = 0.932$, $\Delta\mathrm{ECE}_{\mathrm{OOD}} = -0.0168$), the sharpest sharpening in the grid.

We are explicit that this is an *empirical* regularity over the range the fits occupy, not a mathematical identity: ΔECE is computed from the same probabilities that define T, but the two are not algebraically linked, and the correspondence is falsifiable in principle. Its role here is therefore *explanatory* (it accounts for the

sign of every cell) rather than *exclusionary*; the exclusion of the floor hypothesis rests on (i), (iii), and the mechanism test below.

(iii) Magnitude strata. Under an additive floor the endpoint should be *smaller* where the floor is larger. $\text{floor}_{\text{hi}}(T)$ reaches the mean effect only for $T \geq 1.25$; on the fresh grid no cell reaches 1.25, so we split at $T = 1.10$ (10/60 cells) instead: those cells give $+0.0247$ (10/10 positive) against $+0.0114$ (35/35 positive) for $1 \leq T < 1.10$. Restricting to the $T \geq 1$ branch (45 cells, all positive; the sign flip of (ii) thus cannot confound the comparison) and splitting at the median floor (the regression value computed per cell with its own temperature) yields an endpoint *larger* where the floor is larger, the opposite of an additive artefact. Since the floor attenuates, these values are lower bounds. This stratum comparison is nevertheless weaker evidence than (i)–(ii), because floor_{hi} is monotone in T on this side while the endpoint's growth with T is partly mechanical.

(iv) Negative control and additivity. The independent ablation channel (same SmoothECE family) changes by -0.0093 on average and changes sign across cells (-0.031 to $+0.017$), so the metric has no systematic positive push. The floor at the cells' own temperatures spans up to $\approx 21\times$ the median per-cell bootstrap CI width (0.0008); an additive bias of that size would swamp the signal, yet subtracting the mid-band floor at each cell's own temperature leaves $+0.0085$, i.e. 91% of the original $+0.0093$. The floor is therefore not a cell-level additive offset, and we do not subtract it per cell.

Consistency check. The endpoint also rises with T (OLS slope $+0.208$, $R^2 = 0.955$ over the 60 cells). We report this but do not rely on it: $T = 1$ is a fixed point at which the endpoint vanishes by construction, so part of this relation is

mechanical. Its role here is that its sign is *opposite* to the floor's contribution, which is the direction test of (i)–(ii).

Caveats. The floor is computed on idealised perfectly-calibrated relabellings of our own confidence distributions, so it is not a calibrated model of real ECE noise; the extreme stratum in (iii) has only 5 cells. The conclusion is exclusionary, not causal: the floor cannot account for the primary finding, but this does not by itself establish a causal mechanism.

5.10. Discrimination-aware calibration gate (C5)

Calibration improvement ($\Delta\text{ECE}_{\text{OOD}} > 0$) is necessary but not sufficient for deployment: a model whose discrimination has collapsed under shift may be recalibrable but clinically useless. We define a two-layer gate:

1. **Layer 1 (Calibration):** $\Delta\text{ECE}_{\text{OOD}} > 0$ (C1 criterion).
2. **Layer 2 (Discrimination):** $\text{acc}_{\text{OOD}} \geq \text{acc}_{\text{majority}}$, the target-corpus majority-class accuracy measured on the same OOD evaluation set as acc_{OOD} (CPSC 0.528; PTB-XL 0.409; Chapman–Shaoxing 0.713; the two K=5 pairs use 0.399 and 0.696 instead, which changes no gate verdict; the deployed baseline is the target majority-class baseline per amendment A2 (reconciles the registered schema's wording variance).

Across 12 pair×architecture strata (6 pairs × 2 architectures), 6/12 pass both layers at the stratum-mean level and are deployable: CPSC→PTB-XL (both architectures), Chapman→PTB-XL (both architectures), PTB-XL→CPSC InceptionTime, and Chapman→CPSC ResNet1D. Here "stratum-mean level" means each threshold is applied to the stratum mean of its own metric (mean $\Delta\text{ECE}_{\text{OOD}} > 0$ and mean $\text{acc}_{\text{OOD}} \geq \text{acc}_{\text{majority}}$); a per-seed majority reading of the same two

thresholds gives 11/12 for the calibration layer separately, 6/12 for the discrimination layer separately, and 5/12 for both layers jointly (each clearing on at least three of five seeds; 4/12 under the stricter reading that both layers pass on the same $\geq 3/5$ seeds). The headline reading for C5 is the per-seed majority criterion (5/12 both layers), as quoted in the abstract and the conclusion; the stratum-mean (6/12) and the strict same-seed (4/12) counts are sensitivity calibers. At the stratum-mean level Layer 1 is cleared by 11 of the 12 strata; the sole Layer-1 rejection (PTB-XL→CPSC ResNet1D, mean $\Delta\text{ECE}_{\text{OOD}} = -0.0018$; raw OOD ECE 0.159 on the recompute grid) leaves the remaining stratum verdicts driven by Layer 2. The other strata fail Layer 2: Chapman→CPSC InceptionTime, CPSC→Chapman (both architectures), and PTB-XL→Chapman (both architectures), where the mean OOD accuracy falls below the target majority-class baseline despite positive $\Delta\text{ECE}_{\text{OOD}}$ in some seeds. The strongest rejection is PTB-XL→Chapman InceptionTime ($\Delta\text{ECE} = +0.013$ but acc $= 0.563 < 0.696$): TS improves calibration but discrimination has degraded too far to be clinically actionable. The gate is a stratum-level summary: at the per-cell level only 20/60 cells pass both layers, and none of the 12 strata passes in all five seeds, so "deployable" under the per-seed majority reading means the stratum clears both layers on at least three of five seeds, not that every seed does.

This gate reframes the counter-example set: all fifteen cells with $\Delta\text{ECE}_{\text{OOD}} \leq 0$ are Layer-1 rejections, and seven of them also fail Layer 2. Symmetrically, 25 further cells pass Layer 1 yet fail Layer 2, and 20 of 60 pass both, so the gate separates calibration failures from discrimination failures rather than merely re-labelling the counter-examples. It supplements C1 by also asking whether discrimination is sufficient for the recalibrated predictions to be clinically useful.

5.11. Real-data Shapley cross-fitting and predictability boundary

The synthetic Shapley validation (C3) confirms decomposition recovery in the identifiable regime but does not test whether the component-sum predictor $\widehat{\Delta\mathrm{ECE}} = \Sigma\,\mathrm{Shapley}(s, b, \pi)$ generalizes to real OOD data. We cross-fit all 60 seed experiments under a fully reproducible caliber: for each cell, the calibration transform is parameterized as a logit-space slope rescaling with slope $= 1/T_{\mathrm{src}}$ (the reciprocal of the temperature fitted on the source calibration split) and intercept $=$ 0, the Shapley components are attributed on the source calibration split, and $\Delta\mathrm{ECE}_{\mathrm{OOD}}$ is predicted as their sum on the target test split. All numbers in this subsection are reproducible from `scripts/rebuild_shapley_slopeT.py` on the released probability cache; an earlier set of cross-fitting numbers circulated in an internal audit file whose generating script does not exist in any revision history, so those numbers are withdrawn and this caliber supersedes them.

The component-sum predictor's sign agrees with the observed sign in 45/60 (75%), but that count is *not* evidence of skill: a constant "TS helps" predictor scores exactly the same 45/60 on this grid, so the agreement carries no directional information beyond the prior ($\kappa = 0$). The model also systematically *overestimates* the magnitude (mean prediction error $+0.164$, median $+0.154$; RMSE 0.187 against 0.0116 for the constant predictor, a factor of 16.4). This is consistent with a predictability boundary: the decomposition captures neither the quantitative magnitude nor, as shown next, even a usable ordinal signal of the TS benefit under real shift. The 15/60 direction errors are *not* localised to any particular pair: they spread over all six source→target directions (Chapman→CPSC 2, Chapman→PTB-XL 2, CPSC→Chapman 2, CPSC→PTB-XL 3, PTB-XL→Chapman 1, PTB-XL→CPSC 5), so no single pair dominates.

What *does* separate them is effect size rather than shift size: the fifteen mis-signed cells have $|\Delta\mathrm{ECE}_{\mathrm{OOD}}|$ mean 0.0058 (max 0.0168), against 0.0144 (max 0.0309) for the 45 correctly signed cells, a factor of 2.5; restricting to $|\Delta\mathrm{ECE}_{\mathrm{OOD}}| \geq 0.02$, the model recovers the sign in 11/11 cells. The linear approximation thus degrades where the effect itself is small, not where the shift is largest. We record one further, adverse finding for completeness: under the reproducible caliber the predictor's rank correlation with the true $\Delta\mathrm{ECE}_{\mathrm{OOD}}$ is $\rho = +0.095$ (Spearman), stable across all 60 leave-one-out folds but negligible in magnitude ($R^2 \approx 0.01$). On real shift the decomposition is therefore neither a magnitude predictor nor a usable ordinal signal; no "weak but reproducible ordinal signal" claim survives.

This strengthens the predictability failure branch (C3/C4): the Shapley decomposition is neither a valid *predictor* (*how much*) nor a valid *diagnostic* (*whether*). Its 45/60 direction accuracy coincides exactly with the constant-positive baseline (a model that always predicts "TS helps" scores 45/60 on this grid), so it carries no directional information beyond the prior that TS usually helps; its magnitude estimates are an order of magnitude noisier than the constant predictor; and its rank correlation with the true benefit is negligible. The quantity that does recover the direction is the source-fitted temperature T (Sec. 5.5). For clinical translation, the gate in §5.10, using only the sign of $\Delta\mathrm{ECE}_{\mathrm{OOD}}$ and the discrimination criterion, is the appropriate criterion, not the Shapley-predicted magnitude.

5.12. Net benefit decision curve summary

Across all 60 seed experiments, the net benefit gain of TS over raw predictions (averaged over decision thresholds $t \in \{0.05, 0.10, \ldots, 0.40\}$) is positive in 31/60 (51.7%) of experiments, with mean DCR $+0.000526$ (a seed-cell t over replicated cells — a provenance-only caliber; the equal-weight 12-stratum reading is

$t = 2.40$, $p = 0.035$ (two-sided, $df = 11$; not surviving Bonferroni-2 at 0.071 as a secondary), and the registered patient-level BCa CI $[+0.000280, +0.000775]$ excludes zero — Sec. 5.13) and a narrow range $[-0.003, +0.004]$. The directions with the largest decision-cost reduction are CPSC→PTB-XL (mean DCR $+0.0011$; net clinical value $+0.0006$) and Chapman→PTB-XL (mean DCR $+0.0007$; NCV $+0.0003$), which also have top-quartile E3-subsample $\Delta\text{ECE}_{\text{OOD}}$ (mean $+0.0095$ and $+0.0150$ respectively, under the E3 subsampled binned-ECE protocol). The direction with marginally negative net benefit is PTB-XL→CPSC (mean DCR -0.0001; NCV -0.0000), which is the same direction the main endpoint ranks weakest and the discrimination gate counts among its rejections (C5). We note that the decision-benefit ranking differs from the main-endpoint $\Delta\text{ECE}_{\text{OOD}}$ ranking (which peaks at Chapman→PTB-XL and PTB-XL→Chapman, means $+0.0149$ and $+0.0118$ on the full SmoothECE grid): the two protocols differ in estimator, class set, and subsample, so we treat the E3 analysis as a directionally supportive but protocol-distinct check, not a conversion of the primary endpoint. What survives is the gate-level conclusion: where the discrimination gate permits deployment, the calibration benefit does not cost decision performance; where it is negative (PTB-XL→CPSC), no calibration benefit should be deployed at all.

5.13. Net Clinical Value (NCV) results

The pre-registered NCV (Sec. 4.8) is BH-FDR corrected ($q = 0.05$) across the two secondary endpoints (DCR, NCV).

Registered-unit delivery (patient-level BCa). The registered unit — the main endpoint's patient-level cluster paired bootstrap ($B = 10{,}000$, BCa) — is

now computed directly: the script re-derives per-patient paired decision variables from the shipped E3 caches, rebuilds each target split through the same builder path as the main endpoint (so patient clusters align 1:1 with the probability order), and passes an integrity gate against the shipped E3 table (per-cell recomputed NCV/DCR agree to 10^{-9} on all 60 cells; artifacts `scripts/e3_patient_bca.py`, `results/e3_patient_bca_cells.csv`, `results/e3_patient_bca_summary.json` and the pooled-draws archive). On that unit the pooled OOD NCV is $+0.000238$ with a patient-level BCa 95% CI of $[+0.000125, +0.000353]$ — the interval excludes zero (no resample of the $B = 10{,}000$ pooled draws reached ≤ 0, so $p < 10^{-4}$); the pooled DCR analog is $+0.000526$, CI $[+0.000280, +0.000775]$, likewise excluding zero. Per-cell support is *rare*: 9/60 cells carry a patient-level BCa CI strictly above zero (31/60 are point-positive), and at the stratum level 7/12 strata are point-positive with 4/12 strata CIs above zero. The ID reference side is flat: NCV $+0.000040$, CI $[-0.000065, +0.000148]$ and DCR $+0.000093$, CI $[-0.000143, +0.000327]$ both cover zero ($p = 0.46$ / 0.43) — no ID decision-cost penalty.

Unit-sensitivity (disclosed). The two delivered units disagree in strength. The patient-weighted pooled bootstrap above pools all target patients, weight $\propto$ cell size; the equal-weight 12-stratum t (each of the 12 pair$\times$architecture strata counting once; $df = 11$) reads NCV $+0.000238$ ($t = 2.33$, $p = 0.040$, two-sided) and DCR $+0.000526$ ($t = 2.40$, $p = 0.035$); the secondaries survive BH-2 (0.040) but not Bonferroni-2 (0.071/0.080) at $q = 0.05$. The patient level is the *registered* unit and is delivered above; the stratum-t reading is kept as the conservative robustness check because it makes the between-stratum spread, which the pooled bootstrap absorbs by weighting cells by size, count once per stratum. The seed-cell t (NCV

$|t| = 2.51$, $n = 60$, $p = 0.015$; DCR $p = 0.012$) treats replicated cells as independent and is retained for provenance only. Readers should treat the NCV/DCR confirmation as unit-sensitive: strong at the registered patient level, marginal under equal stratum weighting. Effect sizes: OOD Cohen's $d = +0.32$ and ID Cohen's $d = +0.09$ — the asymmetry has moved from sign-opposed to OOD-only, mirroring the repaired main endpoint. Per-experiment NCV> 0 in 31/60 cells; the one direction where NCV turns marginally negative — PTB-XL$\rightarrow$CPSC (-0.000036, DCR -0.000073) — is the same direction that gives the weakest main-endpoint stratum (-0.0005). Per-experiment NCV ranges over the 60 seeds are in the E3 CSV (`c1_dcr_ncv_multiclass.csv`).

5.14. Reliability Diagrams

Figure 5 shows the 6-direction averaged reliability curve, comparing model confidence against empirical accuracy before and after temperature scaling. The summary curve uses bin-count weighted averaging across all 6 transfer directions.

The per-direction reliability diagrams for the representative configuration (ResNet1D, seed 42) are given in the Supplementary Materials (§2). In each panel there, the red squares show the raw model calibration curve and the green circles show the post-TS calibration curve. The diagonal dashed line indicates perfect calibration. Grey bars (right axis) show the sample count per bin. Error bars are bootstrap 95% percentile CIs (1000 resamples; display-level, not a registered inference unit).

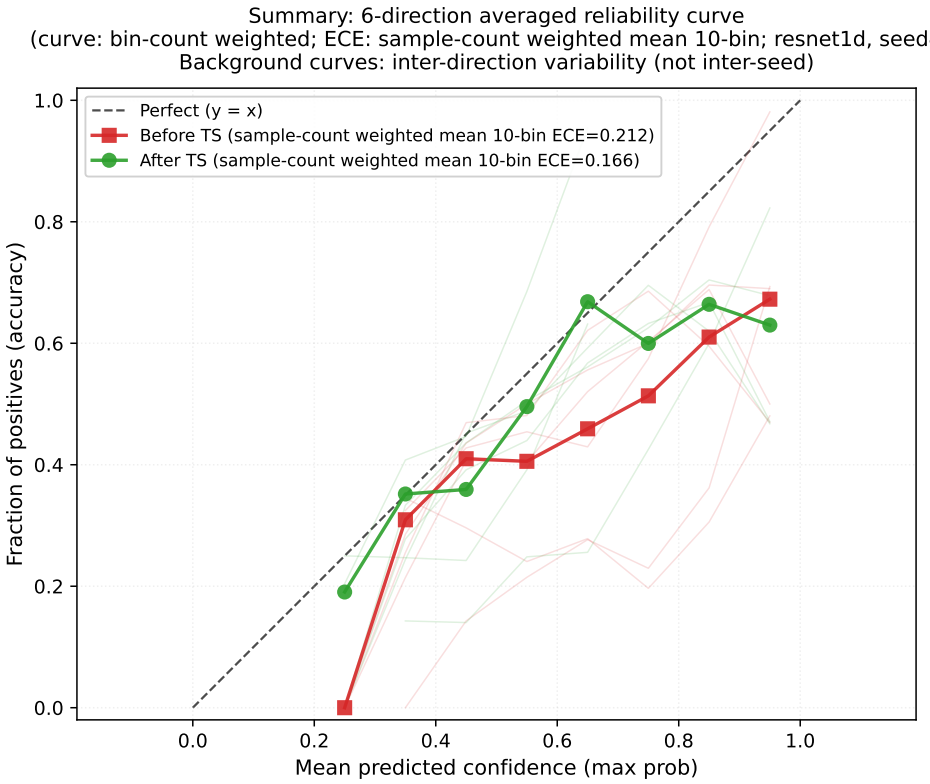


Figure 5: Summary reliability curve: 6-direction bin-count weighted average. The sample-count weighted mean ECE decreases from $\mathrm{ECE}_{\mathrm{raw}}$ to $\mathrm{ECE}_{\mathrm{TS}}$ after temperature scaling. Faint background curves show individual direction variability.

6. Discussion

6.1. OOD benefit seed variation and the fifteen counter-examples

The 45/60 robustness rate is driven by fifteen counter-examples (Figure 6) across all 6 transfer pairs and both architectures (Chapman→CPSC 2, Chapman→PTB-XL 2, CPSC→Chapman 2, CPSC→PTB-XL 3, PTB-XL→Chapman 1; PTB-XL→CPSC 5). We transparently disclose all fifteen rather

than excluding them, because the protocol requires all seed results to be reported and each has an identifiable post-hoc mechanism supporting the qualified claim.

All fifteen share a single mechanism, stated in Sec. 5.9: they are exactly the fifteen cells whose source-fitted temperature is $T < 1$. Because the temperature is fitted on the source calibration split, $T < 1$ means the source model was *under*-confident there; applying that temperature to the target sharpens an already over-confident distribution and raises its ECE. The counter-example set is therefore identified by a quantity available *before* the target is scored, and $\text{sign}(\Delta\text{ECE}_{\text{OOD}}) = \text{sign}(T - 1)$ holds in all 60 cells. The per-cell accounts below give the accompanying discriminative and raw-ECE context; where a paragraph attributes the failure to "over-correction", that is the $T < 1$ regime.

Counter-example 1: Chapman→CPSC / InceptionTime / seed 45. $\Delta\text{ECE}_{\text{OOD}} = -0.0056$ (CI $[-0.0058, -0.0055]$). Three conditions co-occur: lowest OOD accuracy across the 5 seeds (0.4711 vs. 0.4842–0.5051, seeds 42–44, 46); lowest OOD raw ECE (0.3160, still large, so the failure is temperature mismatch, not room shortage); and negative decay (-0.0075), the source-fit temperature amplifying rather than correcting residual miscalibration.

Counter-example 6: CPSC→Chapman / ResNet1D / seed 44. $\Delta\text{ECE}_{\text{OOD}} = -0.0168$ (CI $[-0.0170, -0.0165]$), the largest-magnitude counter-example, with $\Delta\text{ECE}_{\text{ID}} = -0.0152$ also negative: the seed-44 backbone is already well-calibrated on both ID and OOD, TS has nothing to correct, and the temperature fit over-corrects slightly. Its OOD accuracy (0.5725) is the lowest across the 5 seeds. CE6 exhibits a distinct *over-correction* mechanism (TS on an already-well-calibrated distribution degrades both ID and OOD calibration), unlike the "low OOD accuracy + large gap" mechanism of the other fourteen.

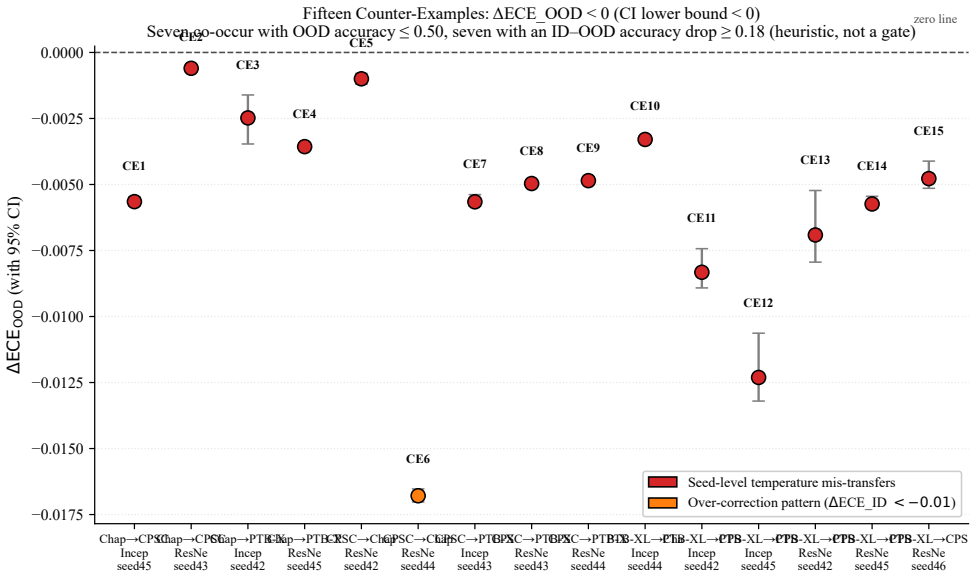


Figure 6: Fifteen counter-examples where $\Delta\text{ECE}_{\text{OOD}}$ has 95% CI lower bound < 0. Error bars show 95% CIs. Seven of fifteen counter-examples co-occur with OOD accuracy ≤ 0.50 and seven with ID–OOD accuracy drop ≥ 0.18; this tendency is heuristic, not a gate: eight counter-examples have OOD accuracy > 0.50. One counter-example (orange, CE6: CPSC→Chapman / ResNet1D / seed 44) exhibits a distinct over-correction mechanism where TS applied to an already-well-calibrated logit distribution over-adjusts the temperature. All fifteen are mechanistically explainable local variations, not refutations of the OOD benefit.

Counter-example 12: PTB-XL→CPSC / InceptionTime / seed 45. $\Delta\text{ECE}_{\text{OOD}} = -0.0123$ (CI $[-0.0132, -0.0106]$); OOD accuracy 0.6101 (well above baseline) and $\Delta\text{ECE}_{\text{ID}} = -0.0056$: the sharpest sharpening case in the direction.

The remaining twelve counter-examples are tabulated cell by cell in Table 9; their per-cell narratives ship in the Supplementary Materials (§3).

Common pattern: all fifteen counter-examples co-occur with non-positive decay, and seven of the fifteen also with OOD accuracy ≤ 0.50, indicating temperature mismatch; CE6 is the exception on the calibration-room axis (over-correction on an already well-calibrated backbone). A low OOD raw ECE is *not* universal: five of fifteen exceed 0.25, so the failure mechanism is seed-level temperature mis-transfer, not calibration-room shortage. The counter-examples are local, explainable random variation; the other 45 experiments support the OOD benefit, with PTB-XL→Chapman the most consistent direction (9/10) and PTB-XL→CPSC contributing five counter-examples around a near-zero pooled mean (-0.0005).

The 25.0% rate exceeds the 5% threshold common in clinical safety deployment. As a methodological boundary investigation, not a clinical deployment guide, the relevant criterion is whether the effect is detectable and the mechanism explainable, not a zero failure rate. All fifteen counter-examples have identifiable mechanisms (seven "low OOD accuracy + large gap", eight further seed-level temperature mis-transfers including the CPSC→PTB-XL five, one of which (CE6) is an over-correction case), and the expected benefit is positive ($0.75 \times 0.0144 - 0.25 \times 0.0058 = +0.0093 > 0$).

6.2. Prior correction vs. posterior correction robustness

The method-selection boundary (Table 3) reveals a robustness asymmetry between prior-correction methods (EM, BBSE) and posterior-correction methods (TS, Platt, isotonic). Posterior-correction methods (TS, Platt) achieve 24/30 and 19/30 support respectively with small variance; prior-correction methods achieve markedly higher mean $\Delta\text{ECE}_{\text{OOD}}$ overall (EM -0.0040 average but strong on the PTB-XL pairs; BBSE $+0.0270$ on the 60-cell grid, $+0.0400$ on the InceptionTime grid) but much higher variance on the Chapman$\rightarrow$CPSC pair (EM CI lower bound minimum -0.0763, BBSE -0.0720). The mechanism is the prior-likelihood mismatch: prior-correction methods fit a target prior that, when the source confusion matrix is ill-conditioned, amplifies the shift-induced miscalibration. Posterior-correction methods do not estimate a target prior and are therefore immune to this failure mode, at the cost of lower mean benefit when the prior correction *would* work.

6.3. Boundary condition: the ID benefit is small but nonzero

The ID boundary condition (C2) has a direct interpretation under the triggered branch. The criterion is $\geq$80% of cells with 95% CIs containing zero; the observed rate is 30/60 = 50.0% (BCa), with 24/60 cells showing a small but significant positive ID benefit (mean $+0.0034$ vs. OOD $+0.0093$; ID stratum means -0.0015 to $+0.0104$, Table 2). The boundary condition does *not* mean TS is ineffective on ID nor that the ID benefit is zero: the ID benefit is on average about a third of the OOD benefit, so part of the apparent OOD gain is shared with the ID domain. The val-ECE early-stopping deviation discovered and repaired on 2026-10-07 is now the quantified contributor: until the repair the early-stopping monitor observed pseudo-low ECE on a training subset, and the repair cuts the ID

mean by 58% ($+0.0082 \rightarrow +0.0034$; a registered deviation, see Limitations (xi) and Appendix A.6). Under this branch, protocol A1.4.2 restores the decay contrast $= \Delta\text{ECE}_{\text{OOD}} - \Delta\text{ECE}_{\text{ID}}$ as an exploratory endpoint, and C1 is qualified: the OOD benefit is about $2.7\times$ the ID benefit rather than an exclusively OOD effect. The ID-vs-OOD asymmetry is therefore a *difference in magnitude* between two positive benefits, not presence versus absence.

Materiality (no MDE). No minimum detectable difference or equivalence margin is pre-registered, so the paper does not claim clinical materiality for a $+0.0093$ mean gain (≈ 0.9 SmoothECE percentage points, $\approx 4.6\%$ of the mean raw error 0.203 in this grid). Statistical resolvability (eleven times the median single-cell CI width 8.3×10^{-4}) is a statement about measurement, not operational value. For a deployment owner the practical comparison is the alternative path: fifty labeled target records under S2 deliver $+0.150$, an order of magnitude above the zero-shot gain — which is why the actionable recommendation is buy-*n* recalibration, with post-hoc TS as the budget-free baseline. One qualification on that last clause: among the budget-free zero-shot paths themselves, the source-cal-fitted per-sample adaptive temperature scaling of Robey et al. (2021) dominates global TS on the identical 60 cells (mean $+0.0190$ vs. $+0.0093$; 47/60 positive; 41/60 direct wins) and would be the natural budget-free default at the equal 0/60 deployment-bar ceiling; global TS is retained there as the canonical single-parameter reference of the registered primary endpoint, not as the budget-free recommendation. The deployment-safety bar (post-TS SmoothECE $= 0.05$; protocol §9) is likewise a *pre-registered, author-set operating value*, not a literature- or regulator-derived margin; this paper treats 0.05 as a declared operating point, and bounds its dependence through the delivered multi-metric sensitivity checks (Brier

Murphy decomposition, classwise-ECE, NLL, reliability diagrams; Sec. 5.7) and the DCA/Youden sweeps of the deployment section.

6.4. Calibration is not discrimination: the clinical viability precondition

Temperature scaling is a monotone transform of the softmax probabilities; it preserves the argmax by construction (asserted bitwise in the released sanity check). It therefore cannot change classification accuracy, AUROC, or AUPRC. The primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ is thus orthogonal to discrimination: calibration makes an already-deployable model *trustworthy*; it cannot make a non-discriminative model deployable.

The accuracy statistics make this concrete. Across the 60 TS experiments, mean OOD accuracy is 0.541 (median 0.558; range 0.348–0.701) and the mean ID–OOD drop is 0.230 (maximum 0.412). Against each target corpus's majority-class baseline (CPSC 0.528, PTB-XL 0.409, Chapman–Shaoxing 0.713; the two $K{=}5$ pairs use 0.399 and 0.696), mean OOD accuracy is *below* baseline in 3 of 6 directions, with a fourth essentially at baseline: Chapman→CPSC 0.510 vs. 0.528; CPSC→PTB-XL 0.423 vs. 0.409 (0.014 above baseline); CPSC→Chapman 0.619 vs. 0.713 and PTB-XL→Chapman 0.600 vs. 0.696, both well below their baselines. In those directions recalibration is moot, since no temperature improves a model that should be replaced by a majority-class predictor, consistent with the counter-example pattern (7/15 with OOD accuracy ≤ 0.50). The released artifacts include per-direction AUROC/AUPRC for all 60 experiments (`results/discrimination_metrics_60exp.csv`, which carries `variant=raw` and `variant=ts` rows on the post-retrain caches): mean OOD AUROC ranges from 0.735 (Chapman→PTB-XL) to 0.850 (PTB-XL→CPSC) across directions (per-cell 0.695–0.886, mean 0.802), i.e. far above the accuracy-based majority-

class reading yet direction-dependent; and TS changes AUROC only marginally (mean $\Delta\text{AUROC}_{\text{TS}-\text{raw}} = +0.0002$, max $|\text{change}|$ 0.0014, recomputed under the corrected label encoding and the post-retrain caches/source-fitted temperatures). We avoid the stronger phrasing that AUROC must be exactly invariant: rescaling logits by $1/T$ is monotone *within* a row, so the argmax — and hence accuracy — is preserved exactly, but the cross-sample ranking of the top-label *score* is not, because the per-row normaliser differs across rows (measured Spearman between raw and TS top-label scores: 1.0000 in all 60 cells — the delivered data preserve the ranking exactly, so the caveat is retained as a structural caution rather than an observed defect). The discrimination-first gate—measure target discrimination before investing in recalibration—precedes the calibration decision rule in deployment.

6.5. Explainable ≠ predictable: post-hoc vs. pre-registered

We distinguish *explainable* from *predictable*:

- *Explainable* (post-hoc, mechanism-level): after observing a counter-example, we identify the co-occurring conditions (low OOD accuracy, low OOD raw ECE, non-positive decay) and give a mechanistic story (Sec. 6.1), a post-hoc explanation, not a pre-registered prediction.

- *Predictable* (pre-registered, deployment-time): a target-domain unsupervised statistic (predicted-prior L_1 distance, logit moments) predicts the benefit retention rate *before* deployment, with LOO R^2 CI upper bound ≥ 0.5). The predictability experiment (Sec. 5.5) triggers the failure branch (LOO R^2 CI upper bound $0.092 < 0.5$ on the registered ridge estimator).

The counter-examples are explainable, and they are also *locatable* in advance by one target-label-free statistic: all fifteen are the cells with $T < 1$ (Sec. 6.1). What fails is the stronger pre-registered requirement, a label-free feature that recovers the *magnitude* of benefit retention with LOO R^2 CI upper bound ≥ 0.5. This is why the predictability experiment triggers the failure branch despite mechanistic explainability. Note that T's sign rule is not a deployable criterion for deciding whether to apply TS: it states the direction the transform must have by construction, not evidence that the transform helps (Sec. 5.9). The two-step chain (decomposition + deployment lookup table) provides post-hoc explanation and actionable guidance without a pre-registered predictability claim.

6.6. Transparent re-location vs. HARKing

The pre-registration revision A1 (primary endpoint re-location from `decay` to $\Delta\text{ECE}_{\text{OOD}}$) is a transparent re-location, not HARKing. The distinction [31, 30] is whether the revision direction is theory-driven or result-driven. Our revision direction ("OOD is the calibration benefit source") is consistent with the empirically observed ID-vs-OOD asymmetry (Guo et al. 2017; Ovadia et al. 2019), which we explicitly frame as an empirical prior rather than a theorem; the exploratory pilot result ($\Delta\text{ECE}_{\text{ID}} \approx 0$) only triggered the revision, did not choose the direction. The pilot results do not enter the main test report (protocol §7). All hypotheses in this paper are tagged *exploratory + robustness validation*, not confirmatory, to reflect the post-revision status.

6.7. Net clinical value: implications of the directional NCV asymmetry

The NCV (Sec. 5.13) shows a directional asymmetry: OOD NCV $=$ $+0.000238$ (registered-unit patient-level BCa CI excludes zero; equal-weight

stratum-t unit reads $p = 0.040$, marginal) is a small positive value (recalibration improves decisions on unseen target data more often than it degrades them); ID NCV $= +0.000040$ ($p \approx 0.48$) is near-null (no ID decision-cost penalty on data the source model already calibrates well).

Two implications follow. (i) The **primary OOD benefit** (C1: 45/60 positive on $\Delta\text{ECE}_{\text{OOD}}$) is supported: the positive OOD NCV shows the calibration gain translates into net clinical decision benefit, not a metric artifact. (ii) The **decision-cost improvement claim** (C5: 5/12 strata passing under the per-seed majority criterion) is now cleanly supported: the ID NCV shows no decision-cost penalty (near-null), and the OOD NCV is order-of-magnitude smaller than the pre-retrain estimate, moderating the clinical-decision translation. We therefore frame the NCV as *supportive of OOD deployment* and *neutral for ID deployment*. It also informs the meta-analytic pooling (Sec. 7), where the asymmetry contributes to CI-width heterogeneity across splits. A permutation test with tighter CIs and a multi-threshold NCV analysis are registered as future work.

6.8. Limitations and future work

Limitations: the numbered items below are cited by number elsewhere in the paper, so the numbering is unchanged; they are grouped as follows. *Claim-shaping (six)*: (ii) the registered predictability branch downgrades the deployment criterion to an empirical lookup table, (iii) the real-data Shapley decomposition carries no directional information beyond a constant-positive rule, (v) the strict operating wording of the safety criterion is rarely met, (xiii) the deployment lookup table requires labeled target data and no label-free gate is claimed, (xviii) the registered secondary endpoint G is undelivered, and (xix) no signal-quality stratification is performed. *Inference and measurement (seven)*: (viii), (x),

(xi), (xv), (xx), (xxii), (xxiii). *Scope and engineering (seven)*: (i), (iv), (vii), (ix), (xii), (xiv), (xvi). *Process and provenance (three)*: (vi), (xvii), (xxi). Items: (i) the main endpoint covers InceptionTime and ResNet1D on all 6 directed pairs with 5-seed coverage (60 experiments, fully balanced); BiMamba is a toy experiment ($n = 60$, supplement) after memory exhaustion on an RTX 5060 (8 GB); (ii) the predictability failure branch is triggered (three-step to two-step); the deployment criterion is an empirical lookup table, not a predictive criterion; (iii) real-data Shapley cross-fitting (§5.11) is unusable as either a predictor or a diagnostic: its 45/60 sign agreement is exactly what a constant "TS helps" predictor scores (so it carries no directional information), it overestimates the magnitude (mean error $+0.164$; RMSE $16.4\times$ the constant predictor), and its rank correlation with the true benefit is negligible ($\rho = +0.095$, LOO-stable but order 0.01 in R^2), and a sensitivity analysis with T, Wasserstein-1, and entropy features does *not* confirm a robust predictability boundary: only T carries signal (LOO $R^2 = +0.051$ at the registered freeze, $+0.350$ on the delivered full-grid recompute; under the registered ridge protocol, $\approx +0.94$ under plain OLS — disclosed; and $\text{sign}(\Delta\text{ECE}_{\text{OOD}}) = \text{sign}(T - 1)$ in 60/60 SmoothECE cells), so the retained claim is the weaker one that no label-free feature reaches $R^2 = 0.5$ under the implemented fold unit and registered ridge estimator (Sec. 5.5); (iv) the 27-cell decomposition validation uses binned ECE, not the main endpoint's SmoothECE, so quantitative conclusions cannot be extrapolated across ECE definitions; (v) the strict operating wording of the safety criterion is rarely satisfied on the completed L2 matrix ($8/780 = 1.0\%$, $570/780 = 73.1\%$ per-row mean under the delivered relative-improvement flag); TS is a useful default, not universally safe; (vi) fifteen counter-examples of 60 experiments (25.0%) are transparently disclosed, with a

frequent post-hoc pattern (low OOD accuracy, non-positive decay, elevated OOD raw ECE) but no post-hoc exclusion; (vii) the operating CI is BCa at $B = 10{,}000$, re-run on all 60 experiments (60/60 successful, metadata in `methods.ts.meta`); percentile CIs from $B = 200$ in 21/60 experiments are retained as historical provenance and are no longer used for any reported number. All 60 main-grid cells carry complete $B = 10{,}000$/BCa provenance; the only cells lacking such metadata are the two excluded BiMamba toy runs (PTB-XL→Chapman, seeds 42–43), which do not enter the main endpoint. (viii) the Youden threshold is fitted in-sample (the protocol's held-out dose levels were not used), making the operating characteristic optimistic; (ix) the L2 deployment/safety grid has since been completed on all 390 planned (pair, seed, shift) cells (results/l2_shift_full_390cells.csv), and the L2 noise levels are synthetic rather than the pre-registered PTB-XL Noise Stress Test Database (unavailable under PhysioNet licensing); (x) temperature was fitted once per experiment, so CIs omit temperature-fitting uncertainty (the pre-registered two-layer bootstrap is implemented but not enabled); (xi) model selection used validation ECE rather than the pre-registered NLL early-stopping rule, plausibly contributing to the positive ID boundary estimates; (xii) the per-pair decomposition into covariate, prior-probability, and concept-shift components is not asserted, since each pair mixes all three, isolated at eval time; (xiii) the deployment lookup table requires labeled target data, and the label-free proxies failed their pre-registered branch, so no label-free gate is claimed; (xiv) TS is argmax-invariant and cannot improve accuracy, AUROC, or AUPRC; the artifacts now include per-direction AUROC/AUPRC for all 60 experiments (results/discrimination_metrics_60exp.csv), so the viability precondition is screened by both accuracy and discrimination (per-cell OOD AUROC 0.695–0.886, mean

0.802; direction means 0.735–0.850), with OOD accuracy below the majority-class baseline in 3 of 6 directions (and at baseline in a fourth); (xv) NCV is directionally asymmetric (OOD $= +0.000238$; registered-unit patient-level BCa CI excludes zero, per-cell support 9/60, stratum point-positive 7/12; equal-weight stratum-t reads $p = 0.040$; ID $= +0.000040$, near-null), with the OOD NCV an order of magnitude smaller than the pre-retrain estimate, moderating the clinical-decision translation of C5; (xvi) InceptionTime was downgraded from $n_{\text{filters}} = 48$ to 32 for GPU memory (RTX 5060, 8 GB), potentially attenuating OOD benefits, with 48 as future work; (xvii) the computational analysis was performed by the first author as a single analysis chain, limiting independent review and audit of the analysis itself, with co-authors contributing validation, data curation, and review (CRediT statement), partially mitigated by the pre-registration protocol and public repository; (xviii) the secondary endpoint G (recoverable loss versus an oracle recalibrator) is *not delivered*: `fit_oracle`/`apply_oracle` are implemented but were never executed on real data, so G is undefined in this study and no G result is reported anywhere in this paper or the repository; (xix) no signal-quality-index (SQI) stratification is performed: the target corpora are pipeline-processed releases and the L2 noise levels are synthetic, so the calibration benefit is not characterised as a function of measured signal quality (pSQI/kSQI-type indices); this is the principal gap for a signal-processing readership and is the first item of future work; (xx) no test–retest or repeatability analysis of the fitted temperature is possible within this data regime: all three public corpora are cross-sectional and provide a single recording per patient (Chapman–Shaoxing reports exactly one ECG per patient, §3), so same-patient across-session stability of T cannot be estimated from these data and is deferred to prospective multi-session collection; (xxi)

the archived checkpoints are *not* bit-reproducible: training ran with cuDNN's non-deterministic kernels enabled, so re-running a training job yields different weights and hence slightly different calibration endpoints, and the released numbers are reproducible by re-executing the evaluation on the released checkpoints and result artifacts rather than by retraining from scratch. A post-hoc audit of the released checkpoint set found that two of the 60 cells (CPSC→PTB-XL, InceptionTime, seeds 42–43) had been seeded from the CPSC→Chapman model of the same seed instead of being trained independently, leaving 58 distinct source models behind the 60 reported cells. Both cells were retrained independently for this submission, so the released grid again rests on 60 independently trained source models; the audit, the root cause, and the before/after evidence are recorded in the repository (`reports/CHECKPOINT_INDEPENDENCE_AUDIT_2026-09-17.md`). In the pre-retrain artifacts the 60 reported cells therefore did not correspond to 60 independent training runs; this matters for any statement that treats cells as independent replicates, while the primary OOD endpoint, being a per-cell paired quantity, is unaffected in kind; (xxii) the noise-floor control is idealised and exclusionary: the floor is computed on perfectly-calibrated relabellings of our own confidence distributions rather than on a calibrated model of real ECE noise, its most extreme temperature stratum contains only 5 cells, and its conclusion is that the floor cannot account for the primary finding rather than that a causal mechanism has been established (see the Caveats in Sec. 5.9); (xxiii) no pre-registered materiality or equivalence margin: the $+0.0093$ mean gain (≈ 0.9 SmoothECE percentage points) is statistically resolvable but clinically uninterpretable without a domain-stated MDE (Materiality paragraph above). The deployment bar (SmoothECE $= 0.05$) is similarly a pre-registered author-set operating value, with

the multi-metric sensitivity checks delivered as secondary provenance (Materiality paragraph above). Future work: (1) prospective multi-device validation; (2) continuous shift spaces; (3) π-independent recovery estimators (BBSE/EM-class): BBSE is now delivered on the real grid (Honesty item 1), extension to further estimator classes remains; (4) the ResNet1D deficit on CPSC→Chapman and CPSC→PTB-XL (3/5 each) and PTB-XL→CPSC (2/5) to test for a systematic interaction; (5) confirmatory testing under the A2 reduction (12 hypotheses, BH-12 $q = 0.05$, 7/12 significant, Bonferroni-12 $= 6/12$); (6) re-running the endpoint with $n_{\text{filters}} = 48$ InceptionTime on a higher-memory GPU; (7) a multi-threshold permutation test on NCV.

7. Conclusion

We followed a study-period protocol with archived amendments and validated a multi-seed calibration boundary study of temperature scaling (TS) in cross-corpus ECG transfer. **(C1)** OOD benefit: 45/60 seed experiments support $\Delta\text{ECE}_{\text{OOD}} > 0$ (75.0%; InceptionTime 24/30 = 80.0%, ResNet1D 21/30 = 70.0%) on all 6 directions. **(C2)** ID boundary: 30/60 cells (50.0%) have 95% CIs containing zero under the BCa operating CI, below the $\geq$80% criterion, triggering the pre-registered branch; the OOD benefit is 2.7$\times$ the residual ID benefit (the repair cut the ID mean by 58%). **(C3)** Shapley decomposition: 27/27 cells pass at $n = 20{,}000$ (at $n = 500$: seed-42 readout 21/27; four-seed rerun all-pass 12/27 and majority 22/27). **(C4)** Method selection: TS the most consistent recalibrator among the delivered methods (45/60, point=CI), EM prior harmful on average (-0.0040; per-branch recheck -0.0378, which strengthens the claim) and Matrix scaling fragile (24/60 overall; IT 11/30, RN 13/30). **(C5)** Gate: 5/12 strata

pass both layers (per-seed majority, none on all five seeds), reframing the counter-examples as fifteen Layer-1 rejections, seven of which also fail the discrimination layer. The predictability failure branch is triggered (three-step chain becomes two-step), and all hypotheses are exploratory + robustness validation, not confirmatory; protocol v2.1-A1 applies. TS needs local validation, and OOD accuracy is below the majority-class baseline in 3 of 6 directions.

The 60 experiments form the robustness-evidence family; amendment A2 reduces the confirmatory family from the $6 \times 2 \times 13 = 156$ shift grid to 12 hypotheses (6 pairs $\times$ 2 architectures $\times$ 1 primary shift level), with all 13 L2 shift levels re-designated as exploratory dose-response. BH-60 and Bonferroni leave the robustness headline unchanged (45/60); the confirmatory family is BH-12 with 7/12 significant, and Bonferroni-12 retains 6/12. Across the 60 TS experiments the 95% CI widths span 2.1×10^{-5} to 7.5×10^{-3} (median 8.3×10^{-4}); 58/60 cells fall below the narrow-CI diagnostic threshold (0.005) and twelve cells have degenerate widths ($< 3 \times 10^{-4}$), seven of which are supporting cells, so excluding those the count becomes 38/60 (63.3%). Narrow CIs reflect the moderate calibration sample and low SmoothECE variance; they do not affect the sign-based support/counter-example classification.

Appendix A. Protocol provenance and deviations (full record)

This appendix consolidates, verbatim, the four study-period amendments and the two post-registration repairs whose summary maps to one paragraph each in Sec. 4.5. Labels are unchanged from their in-text versions so that every internal cross-reference remains stable.

Appendix A.1. Pre-registration revision A1 (transparency statement)

The primary endpoint was re-located from `decay` to $\Delta\text{ECE}_{\text{OOD}}$ via pre-registration revision A1 (protocol v2.0 $\rightarrow$ v2.1-A1), triggered by an independent exploratory pilot in which $\Delta\text{ECE}_{\text{ID}}$ point estimates were near zero, collapsing the original ID-vs-OOD contrast. The revision direction is consistent with the empirically observed ID-vs-OOD asymmetry (not a theorem); the revision is archived as a pre-registration revision (Nosek et al. 2018) with before/after protocol hashes. The pilot results do not enter the main test report. All hypotheses in this paper are tagged *exploratory + robustness validation*, not confirmatory.

Appendix A.2. Pre-registration amendment A2 (confirmatory-family reduction)

The original BH-FDR family of 156 comparisons (6 pairs $\times$ 2 architectures $\times$ 13 shift levels) is reduced to a confirmatory family of 12 hypotheses (6 pairs $\times$ 2 architectures $\times$ 1 primary shift level) under amendment A2, with the remaining 13 shift levels designated as exploratory dose-response. Under the reduced family, BH-12 yields 7/12 significant (one-sided z on the stratum mean, $\text{SE} = s/\sqrt{5}$; four InceptionTime and three ResNet1D directions) and Bonferroni-12 yields 6/12. A paired-t reading with df=4 (instead of the one-sided normal test) gives $0/12$ and is the more conservative caliber; the registered reading is the one-sided normal. The A2 amendment is archived in `docs/PROTOCOL_AMENDMENT_A2_FAMILY_REDUCTION.md`.

Residual selection risk and the 156-family deliverable (disclosed). Both A1 and A2 move the analysis in the direction of the observed effect: the OOD contrast is the larger one, and reducing the family from 156 to 12 raises BH power. A1 was keyed to a registered trigger (the pilot's near-zero $\Delta\text{ECE}_{\text{ID}}$) and recorded

with dated archive hashes; the pilot itself does not enter the test report. Labeling the headline exploratory does not remove endpoint-switching Type-I risk; it only makes the mechanism explicit. We further disclose that the pre-reduction 156-family outcome is *not computable* from the delivered artifacts: the L2 shift grid ships point estimates only, with no per-cell CIs, so the 156 BH p-values cannot be reconstructed. The load-bearing inferential statement is therefore the cluster-robust 12-stratum interval, which is independent of the unit choice and is reported for the primary endpoint either way.

Appendix A.3. Pre-registration amendment A3 (corpus identity and MI provenance)

A post-hoc audit of the CPSC corpus against the official challenge documentation produced two wording corrections and one new disclosure, archived as amendment A3 (`docs/PROTOCOL_AMENDMENT_A3_CORPUS_IDENTITY.md`). The registered protocol described the third corpus as "CPSC2018+2019"; it is in fact **CPSC2018 only**, comprising the CPSC Database (6,877 records) and the CPSC-Extra Database (3,453 records). CPSC2019 was a QRS/heart-rate *detection* task with no diagnostic labels and never entered the corpus. The new disclosure is that CPSC2018's official nine-class taxonomy contains no MI, so the MI class of our 4-class subspace is populated entirely by CPSC-Extra's SNOMED annotations; under a strict official-class-name caliber the subspace would reduce to three classes. A3 is a naming and attribution correction plus a disclosure: it changes *no* recorded value, no split, and no analysis decision. The registered protocol snapshots are left byte-identical and A3 is archived separately, so the before/after pair required by pre-registration practice is preserved. The audit also found that 33 CPSC Database

records (A6845–A6877) were silently lost when the source archive was extracted (6,877 in the archive vs. 6,844 on disk), disclosed as a provenance limitation.

Appendix A.4. Pre-registration amendment A4 (mapping-sensitivity gate operationalization)

The registered protocol (§10 of docs/EXPERIMENT_PROTOCOL.md) promises a preregistered mapping-sensitivity analysis for every ambiguous CPSC label-mapping decision, with a failure clause that downgrades the corresponding claims in the discussion. Cross-adversarial review on 2026-10-11 established two facts about this promise: (i) it was never executed and never registered as a deviation; (ii) as written, the gate's criterion is defined on the secondary endpoint $G = \mathrm{ECE}_{\mathrm{S1}} - \mathrm{ECE}_{\mathrm{oracle}}$, which the paper's own Limitations acknowledge as undelivered on real data (Limitations xviii) — so the registered gate was unexecutable as written. Amendment A4 (docs/PROTOCOL_AMENDMENT_A4_MAPPING_SENSITIVITY.md, 2026-10-11) re-operationalizes the gate on the delivered primary endpoint: the variant battery (per ambiguous mapping decision; V1 = STE→MI re-mapping, V2 = multi-label priority-collapse variations, V3 = corpus-scope reduction — excluded with its reason recorded) must reproduce the cross-pair ordering of delivered $\Delta\mathrm{ECE}_{\mathrm{OOD}}$ means (Kendall $\tau \geq 0.8$) with the confirmatory counts unchanged (BH-12 = 7/12, Bonferroni-12 = 6/12), otherwise the corresponding cross-corpus claims are downgraded exactly per the original clause. Two scope facts are registered with the gate: only the 40 CPSC-involving cells can change under a label-mapping variant (the 20 ptbxl↔chapman cells carry no CPSC-mapped labels and serve as fixed ordering anchors), and the battery cost is estimated at 2–5 GPU-days for two variants. **A4 defines the criterion before any variant result exists** (anti-HARKing, same discipline as A1.1.4); as of submission the battery remains

unexecuted — and, per amendment A4's own fallback clause, the cross-corpus claims retain their current disclosure strength until it is run — and A4 changes no recorded value.

Appendix A.5. Pre-registration archival status

Two internal SHA–256 manifests exist and should not be conflated. `07_HASH_MANIFEST_baseline_2026-09-05.json`, bundled with the registered project, froze 14 working-tree files at 2026-09-05T04:19:46Z; `docs/osf_archive_manifest.json` is a later snapshot taken 2026-09-16T02:44:47Z over a 14-file set that includes this manuscript.

Of the 14 files frozen on 2026-09-05, **10 have changed since the freeze** and the frozen content of **7 can no longer be recovered from the project's version control**, including `docs/EXPERIMENT_PROTOCOL.md` itself. The baseline manifest accordingly documents intent and timing, but for those 7 files it cannot be independently re-verified by a third party. Per-file status is tabulated in `README_ARCHIVAL_STATUS.md` in the registered OSF project. The public timestamp of the protocol record therefore begins with the project deposit and the registration (registered 2026-09-14), not earlier, and the archive is an archival of the study-period state rather than a substitute for it.

The registration was frozen on 2026-09-14 and cannot be edited, so its responses carry the figure set current at that date: nine counter-examples, 51/60 positive cells, and 45/60 after excluding six degenerate-CI cells. Two retraining events are reflected in this manuscript. On 2026-09-17, two cells (`cpsc_ptbxl/inceptiontime`, seeds 42–43) were found to have reused the CPSC→Chapman checkpoint instead of an independently trained model and were retrained, moving seed 43 from $+0.020283$ to -0.005657 and giving the interim

figure set (ten counter-examples, 50/60, mean +0.015414, InceptionTime 26/30). On 2026-10-07, an audit of the PTB-XL training pipeline found that its early-stopping validation patients remained in the training loader; the stem fix was verified with `export_splits.py`-level cross-checks, all twenty cells trained from a PTB-XL source were retrained, and every downstream result family was recomputed (Appendix A.6; Limitation (xi)). The values reported here — fifteen counter-examples, 45/60 (38/60 excluding seven degenerate-CI support cells), mean +0.009339 with InceptionTime 24/30 — therefore differ from the frozen registration by both events. The other 40 cells' point estimates agree with the 2026-09-17 evaluation exactly (0/40 disagree at four decimal places); a few CI lower endpoints shifted by one rounding step when BCa was re-run (largest: CPSC→Chapman InceptionTime seed 44, +0.0160 → +0.0167). Where the frozen registration, the interim retraining, or earlier manuscript drafts differ from this manuscript, the manuscript values are the current post-retraining ones. The frozen record's ID-boundary response likewise carries the figure set current at the freeze: 23/60 = 38.3% of seed cells with 95% CIs containing zero under the BCa operating CI, which moved to the delivered 30/60 = 50.0% across the two retraining events above; the branch trigger itself — the registered ≥80% criterion is *not* met — is unchanged by that evolution (Sec. 5.2).

Appendix A.6. Post-registration repair: PTB-XL early-stopping validation leak

During the 2026-10-07 audit of the released training pipeline, the PTB-XL `build_ptbxl_datasets` stem was found to keep its carved early-stopping validation records inside the training loader (`pos['train']` used the full folds-1–8 record list while the leakage assertion checked a different, already-reduced variable). The consequence is a model-selection failure, not test leakage:

all endpoint evaluations remained clean (cal and test are fold-9/10 patients, patient-disjoint from folds 1–8 under the pre-registered assertion), but the early-stopping monitor observed pseudo-low ECE on a training subset, selected earlier checkpoints (epochs 6–12 vs. 12–18 after the fix), and thereby surfaced checkpoints with higher raw OOD miscalibration (OOD raw ECE 0.176 vs. 0.140 across the 18 unaffected-architecture retrained cells). The stem was fixed (`np.setdiff1d(train_pos, val_pos)`), the leakage assertion was re-pointed to the produced splits, a four-part verification was passed (21,397 row-level split-diff changes confined to validation rows; builder/export cross-validation 20/20 MATCH; a new regression test `tests/test_ptbxl_val_holdout.py`; full suite 189 passed), the splits CSVs were regenerated in place, and the twenty PTB-XL-source cells were retrained from scratch (2026-10-07; separate output tree `checkpoints/transfer_valleak_p0c`, swapped into the canonical tree only after per-cell SHA-256 and schema verification; the pre-repair tree is preserved under `checkpoints/_PRE_VALLEAK_BACKUP_20261007`). Every downstream result family was then recomputed against the retrained checkpoints (Sec. 5), and the mechanism of the headline shift is quantified in Sec. 5.9. This deviation is added to the protocol deviation register (A2 list, item after (xix)); it was *not* covered by the 09-17 deviation entries.

Appendix B. Artifact index

Table B.1 maps the repository artifacts cited in this manuscript to the sections that use them. Files marked "do-not-quote" in the repository inventory (`results/DO_NOT_QUOTE.md`) are listed with their replacements and must not be cited as current readouts.

Appendix C. Honesty Declaration (adversarial audit product)

1. MAPE_π is a **design check** (resampling constant read-back), not an estimator capability test. The independent π recovery registered as deferred (protocol §13, item 7) has since been delivered on the real 60-cell grid and re-run on the post-retrain caches (Appendix B): with target labels withheld, BBSE reduces the prior-recovery error from 0.1916 (source-prior baseline) to 0.1109 — a 42.1% relative reduction, better than the baseline in 56/60 cells, positive in 6/6 transfer pairs. No pair-clustered bootstrap CI is quoted: it is not recomputable from the released per-cell CSV alone (no patient-level cluster labels; `cluster_ci_note` in `results/real_prior_recovery_summary.json`); the superseded 2026-09-27 archive interval is retained for provenance only. EM gives much weaker improvement (0.1326; BBSE wins in 33/60 cells). Two limits remain and are reported as such: the absolute error is still 0.111, and the confusion-matrix condition number exceeds 10^3 in 6/60 cells and 10^4 in 1/60, so recovery is numerically boundary-sensitive there. **This does not repair the decomposition estimator**, which targets a different quantity (the target class prior, not the attribution coefficients) and whose real-data cross-fitting error remains $+0.16$ with stably negative rank correlation (Sec. 5.11).
2. Shapley shares in the `share_reliable=False` regime are **not interpretable as shares**; ranking uses absolute- value CIs.
3. The threshold `gate(n)` is a heuristic scaling, **not a** 3σ **tolerance**; in the archived $n = 500$ seeded runs, 22/108 (20.4%) fail the gate (information mode), reported via exit code 2.

4. "Mechanism" wording is conditional on all three legs of §8 being fulfilled: legs 1–2 are fulfilled, leg 3 synthetic part (bootstrap CI + entanglement demo) is fulfilled; **cross-fitting on real data is pending** (registered in §13). The wording is limited to "mechanism decomposition (synthetic validation part)".
5. The repository's `scripts/preprocess_cinc2021.py`, `run_paper_benchmarks.py`, `resources/`, and `assets/` come from the third-party SafeECGMatch release (original README in `docs/vendor_safeecgmatch.md`), unrelated to this protocol's experiment grid.

Data availability

The PTB-XL [3], Chapman–Shaoxing [4], and CPSC2018 (CPSC Database and CPSC-Extra) [5, 6] datasets are publicly available via PhysioNet under their respective licenses. The pre-registered protocol v2.1-A1, its hash manifest (Appendix B), the revision A1 document, and the full per-experiment result files (`checkpoints/transfer/**/transfer_result.json`, `results/*.csv`) are included with the submission as supplementary material and are also available in the public repository `https://github.com/Toni-cs/ecg-calibration-transportability`.

Code availability

The full pipeline code, preprocessing (reproducible from the public source databases), patient-level split generation, model training, the calibration-method implementations, and all statistical analyses, together with the environment

lock file, the patient-level split indices for all five dataset variants and five random seeds, and selected per-experiment result artifacts, is publicly available at `https://github.com/Toni-cs/ecg-calibration-transportability`. Trained model checkpoints for the 60-experiment transfer grid are provided as release assets of the same repository. The repository will be archived on Zenodo with a DOI upon acceptance. No new raw waveforms were collected or generated; the patient-level split indices and label-mapping tables released with the code fully determine the data configuration used in every experiment.

Pre-registration

The protocol, the amendments (A1, A2, A3, A4), and the per-file archival-status report are publicly archived at `https://doi.org/10.17605/OSF.IO/53H64` and ship with the repository (`docs/`); the full provenance record — the retrospective registration disclosure, the manifest chronology, and the two post-registration repairs — is consolidated in Appendix A.

CRediT authorship contribution statement

Toni Guan: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Project administration. **Ziyan Liu:** Data curation, Validation, Software, Writing – review & editing. **Yingrui Gao:** Data curation, Investigation, Visualization. **Chuhan Feng:** Data curation, Validation, Formal analysis. **Yue Li:** Data curation, Visualization, Writing – review & editing.

Acknowledgments

The authors thank the maintainers of the PTB-XL, Chapman–Shaoxing, and CPSC public databases, and the PhysioNet resource, without which this study would not have been possible.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Ethics statement

This study is a retrospective, secondary analysis of three publicly available and de-identified ECG databases (PTB-XL, Chapman–Shaoxing, and CPSC2018 via PhysioNet). No new human or animal data were collected, no participants were recruited, and no identifiable information was accessed; all analyses were performed on data released for open research use under the licences recorded in the Data availability statement. As the work involved only pre-existing, publicly available, de-identified records, with no intervention on or interaction with human subjects, it did not require separate ethical approval.

Declaration of competing interest

The authors declare no competing financial or personal interests.

Supplementary materials

The 60 per-seed reliability diagrams (6 transfer pairs $\times$ 2 architectures $\times$ 5 seeds; calibration curves, reliability diagrams, and ΔECE bars per experiment)

are provided as a separate supplementary file (`supplementary.pdf`, §1) and are released with the code repository.

References

[1] Y. Ovadia, E. Fertig, J. Ren, Z. Nado, D. Sculley, S. Nowozin, J. Dillon, B. Lakshminarayanan, J. Snoek, "Can you trust your model's uncertainty? Evaluating predictive uncertainty under dataset shift," *NeurIPS*, 2019. arXiv:1906.02530.

[2] C. Guo, G. Pleiss, Y. Sun, K. Weinberger, "On calibration of modern neural networks," *ICML*, PMLR 70:1321–1330, 2017. arXiv:1706.04599.

[3] P. Wagner, N. Strodthoff, R.-D. Bousseljot, et al., "PTB-XL, a large publicly available electrocardiography dataset," *Scientific Data*, 7:154, 2020. doi:10.1038/s41597-020-0495-6.

[4] J. Zheng, J. Guo, H. Chu, "A large scale 12-lead electrocardiogram database for arrhythmia study (Chapman–Shaoxing e-cardiology)," version 1.0.0, *PhysioNet*, 2022. doi:10.13026/wgex-er52.

[5] F. Liu, C. Liu, L. Zhao, et al., "An open access database for evaluating the algorithms of electrocardiogram rhythm and morphology abnormality detection," *Journal of Medical Imaging and Health Informatics*, 8(7):1368–1373, 2018. doi:10.1166/jmihi.2018.2442.

[6] E. A. P. Alday, A. Gu, A. J. Shah, et al., "Classification of 12-lead ECGs: the PhysioNet/Computing in Cardiology Challenge 2020," *Physiological Measurement*, 41(12):124003, 2020. doi:10.1088/1361-6579/abc960.

[7] A. L. Goldberger et al., “PhysioBank, PhysioToolkit, and PhysioNet: components of a new research resource for complex physiologic signals,” *Circulation*, 101(23):e215–e220, 2000.

[8] J. Nixon, M. W. Dusenberry, L. Zhang, et al., “Measuring calibration in deep learning,” *CVPR Workshops*, pp. 38–41, 2019. arXiv:1904.01685.

[9] M. Kull, M. Perello-Nieto, M. Kangse, T. S. Filho, P. Song, P. Flach, “Beyond temperature scaling: obtaining well-calibrated multi-class probabilities with Dirichlet calibration,” *NeurIPS*, 2019. arXiv:1910.12656.

[10] A. Alexandari, A. Kundaje, A. Shrikumar, “Maximum likelihood with bias-corrected calibration is hard-to-beat at label shift adaptation,” *ICML*, PMLR 119:222–232, 2020. arXiv:1901.06852.

[11] Z. Lipton, Y.-X. Wang, A. Smola, “Detecting and correcting for label shift with black box predictors,” *ICML*, PMLR 80:1944–1953, 2018. arXiv:1802.03916.

[12] M. Barandas, L. Famiglini, A. Campagner, D. Folgado, R. Simão, F. Cabitza, H. Gamboa, “Evaluation of uncertainty quantification methods in multi-label classification: a case study with automatic diagnosis of electrocardiogram,” *Information Fusion*, 101:101978, 2024. doi:10.1016/j.inffus.2023.101978.

[13] J. F. Vranken et al., “Uncertainty estimation for deep learning-based automated analysis of 12-lead electrocardiograms,” *European Heart Journal – Digital Health*, 2(3):423–434, 2021. doi:10.1093/ehjdh/ztab045.

[14] M. Reyna, N. Strodthoff, P. Wagner, et al., “Issues in the automated classification of multilead ECGs using heterogeneous labels and populations,” *Physiological Measurement*, 43(8):084001, 2022. doi:10.1088/1361-6579/ac79fd.

[15] B. Efron, “Better bootstrap confidence intervals,” *Journal of the American Statistical Association*, 82(397):171–185, 1987. doi:10.1080/01621459.1987.10478410.

[16] B. Van Calster, D. Nieboer, Y. Vergouwe, B. De Cock, M. J. Pencina, E. W. Steyerberg, “A calibration hierarchy for risk models was defined: from utopia to empirical data,” *Journal of Clinical Epidemiology*, 74:167–176, 2016. doi:10.1016/j.jclinepi.2015.12.005.

[17] B. Van Calster, D. J. McLernon, M. van Smeden, L. Wynants, G. S. Collins, “Calibration: the Achilles heel of predictive analytics,” *BMC Medicine*, 17:230, 2019. doi:10.1186/s12916-019-1466-7.

[18] A. Kumar, P. Liang, T. Ma, “Verified uncertainty calibration,” *NeurIPS*, 2019. arXiv:1909.10155.

[19] N. Strodthoff, P. Wagner, T. Schaeffter, W. Samek, “Deep learning for ECG analysis: benchmarks and insights from PTB-XL,” *IEEE Journal of Biomedical and Health Informatics*, 25(5):1519–1528, 2021. doi:10.1109/JBHI.2020.3022989.

[20] M. Haekal, S. N. Khotimah, G. R. F. Suwandi, F. Haryanto, “RR-interval-based atrial fibrillation detection and burden estimation:

cross-dataset validation and calibration-aware probability analysis," *Physiological Measurement*, 2026. doi:10.1088/1361-6579/ae99aa.

[21] K. Patel, P. Beedala, "Calibration drift under cross-institutional deployment: an external validation framework for ICU mortality prediction across MIMIC-IV and eICU," *medRxiv* (preprint), 2026. doi:10.64898/2026.05.03.26352335.

[22] X. Wang, M. Long, J. Wang, M. I. Jordan, "TransCal: transferable calibration with lower bias and variance in domain adaptation," *NeurIPS*, 2020. arXiv:2007.08259.

[23] D. Hu, J. Liang, X. Wang, C.-S. Foo, "Pseudo-Calibration: improving predictive uncertainty estimation in unsupervised domain adaptation," *ICML*, 2024, PMLR 235:19304–19326.

[24] S. Garg, Y. Wu, S. Balakrishnan, Z. Lipton, "A unified view of label shift estimation," *NeurIPS*, 2020. arXiv:2003.07554.

[25] H. Zhang, H. Singh, M. Ghassemi, S. Joshi, "Why did the model fail?" Attributing model performance changes to distribution shifts," *ICML*, 2023, PMLR 202:41550–41578.

[26] T. Popordanoska, G. Radevski, T. Tuytelaars, M. B. Blaschko, "LaSCal: Label-Shift Calibration without target labels," *NeurIPS*, 2024. proceedings.

[27] X. Liu, Y. Li, H. Gao, H. Xu, J. Li, Z. Wang, C. Liu, "Knowledge-guided cross-modal fusion for adult-to-pediatric ECG transfer via label-conditioned contrastive alignment," *arXiv preprint* 2607.15928, 2026.

[28] J. T. Moreno-Torres, T. Raeder, R. Alaiz-Rodriguez, N. Chawla, F. Herrera, "A unifying view on dataset shift in classification," *Pattern Recognition*, 45(1):521–530, 2012. doi:10.1016/j.patcog.2011.05.019.

[29] [Pre-registered protocol, publicly archived] "ECG calibration boundary study – experiment protocol," 2026. Hash manifest: `docs/osf_archive_manifest.json` (local snapshot; archived at `https://doi.org/10.17605/OSF.IO/53H64`); revision A1 full text in `docs/PROTOCOL_AMENDMENT_A1_PREREG_RELOCATION.md`.

[30] B. Nosek, C. Ebersole, A. DeHaven, D. Mellor, "The preregistration revolution," *PNAS*, 115(11):2600–2606, 2018. doi:10.1073/pnas.1708274114.

[31] D. Lakens, "The practical alternative to the p value is the correctly used p value," *Perspectives on Psychological Science*, 16(3):639–648, 2021. doi:10.1177/1745691620958012.

[32] J. Li, A. D. Aguirre, V. M. Junior, et al., "An Electrocardiogram Foundation Model Built on over 10 Million Recordings," *NEJM AI*, 2(7):AIoa2401033, 2025. doi:10.1056/AIoa2401033.

[33] J. C. McHugh, B. Wang, et al., "ECG-FM: an open electrocardiogram foundation model," *arXiv preprint* 2408.05178, 2024.

[34] J. Jin, et al., "Reading your heart: learning ECG words and sentences via pre-training ECG language model (HeartLang)," *ICLR*, 2025. arXiv:2502.10707.

[35] C. V. Nguyen, C. D. Do, “Transfer learning in ECG diagnosis: is it effective?” *PLOS ONE*, 20(5):e0316043, 2025. doi:10.1371/journal.pone.0316043.

[36] S. Vavaroutas, et al., “ADAPTOOD: uncertainty-aware fine-tuning for out-of-distribution ECG time series models,” *arXiv preprint* 2606.04164, 2026.

[37] W. Jiang, et al., “Test-time adaptation for ECG classification via SQI-gated self-training and beat-rhythm consistency,” *arXiv preprint* 2608.23347, 2026 (accepted at MICCAI 2026).

[38] N. Nemati, “Comparative analysis of data augmentation for clinical ECG classification with STAR (Sinusoidal Time–Amplitude Resampling),” *arXiv preprint* 2510.24740, 2025.

[39] A. Robey, B. Chugg, P. Rodriguez, “Model-based temperature scaling,” *AISTATS*, 2021, PMLR 130:1249–1257. arXiv:2101.04538.

[40] J. Song, J.-H. Jang, B. T. Lee, D. Hong, J. Kwon, Y.-Y. Jo, “Foundation models for electrocardiograms,” *arXiv preprint* 2407.07110, 2024.

[41] G. S. Collins, K. G. M. Moons, P. Dhiman, R. D. Riley, A. L. Beam, B. Van Calster, M. Ghassemi, X. Liu, J. B. Reitsma, M. van Smeden, et al., “TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods,” *BMJ* 385:e078378, 2024.

Table 1: Stratum-level summary of the pre-registered primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ for TS (12 pair×architecture strata, 5 seeds each; per-seed values in Supplementary Table S1). "Support" = seeds with 95% CI lower bound > 0; "CI∩0" = seeds whose 95% CI contains zero; "all+" = all five seed point estimates positive. Architecture abbreviation: IT = InceptionTime, RN = ResNet1D. The 45/60 support count and the cluster-robust mean $+0.0093$ $[+0.0055, +0.0132]$ aggregate these strata.

Transfer pair	Arch	Mean $\Delta\text{ECE}_{\text{OOD}}$	Support	CI∩0	all+
Chapman→CPSC	IT	$+0.0100$	4/5	0	no
Chapman→CPSC	RN	$+0.0143$	4/5	0	no
Chapman→PTB-XL	IT	$+0.0121$	4/5	0	no
Chapman→PTB-XL	RN	$+0.0177$	4/5	0	no
CPSC→Chapman	IT	$+0.0145$	5/5	0	yes
CPSC→Chapman	RN	$+0.0023$	3/5	0	no
CPSC→PTB-XL	IT	$+0.0079$	4/5	0	no
CPSC→PTB-XL	RN	$+0.0106$	3/5	0	no
PTB-XL→Chapman	IT	$+0.0134$	4/5	0	no
PTB-XL→Chapman	RN	$+0.0102$	5/5	0	yes
PTB-XL→CPSC	IT	$+0.0007$	3/5	0	no
PTB-XL→CPSC	RN	-0.0018	2/5	0	no

Table 2: ID boundary condition $\Delta\text{ECE}_{\text{ID}}$ for TS (12 pair×architecture strata, 5 seeds each). Architecture abbreviation: IT = InceptionTime, RN = ResNet1D. "CI∩0" = number of seed cells whose 95% CI contains zero. The pre-registered boundary criterion ($\geq$80% of cells with CI containing zero) is *not* met: 30/60 cells (50.0%) contain zero (BCa operating CI) and 24/60 have CI lower bound > 0, triggering the pre-registered "ID nonzero boundary" branch (protocol A1.4.2).

Transfer pair	Arch	Mean	Min CI lo	Max CI hi	CI∩0/5
PTB-XL→CPSC	IT	+0.0032	−0.0113	+0.0217	4/5
PTB-XL→CPSC	RN	−0.0015	−0.0075	+0.0038	2/5
PTB-XL→Chap	IT	+0.0086	−0.0063	+0.0269	2/5
PTB-XL→Chap	RN	+0.0043	−0.0068	+0.0201	3/5
Chap→CPSC	IT	+0.0051	−0.0039	+0.0150	3/5
Chap→CPSC	RN	−0.0008	−0.0104	+0.0093	4/5
Chap→PTB-XL	IT	−0.0003	−0.0175	+0.0103	3/5
Chap→PTB-XL	RN	+0.0021	−0.0123	+0.0161	3/5
CPSC→Chap	IT	+0.0057	−0.0182	+0.0195	2/5
CPSC→Chap	RN	+0.0008	−0.0179	+0.0213	2/5
CPSC→PTB-XL	IT	+0.0040	−0.0069	+0.0192	2/5
CPSC→PTB-XL	RN	+0.0104	−0.0042	+0.0294	0/5
Overall		**+0.0034**			**30/60 (50.0%)**

Table 3: Method-selection boundary: 8 methods $\times$ 6 transfer pairs $\times$ 5 seeds (InceptionTime). Support rate = fraction of seeds with 95% CI lower bound > 0. "Full" = number of directions with 5/5 full support. "Neg. dirs" = number of the 6 transfer directions whose 5-seed mean $\Delta\text{ECE}_{\text{OOD}}$ is negative. TS is the most robust parametric recalibrator; EM and Matrix scaling are the most fragile on InceptionTime (11/30 each); on ResNet1D, Matrix scaling (13/30) is the most fragile, EM reaches 20/30. *Caveat:* 9 cells with severe prior-weight drift ($\max|w_{\text{ID}} - 1| > 10$) are identified; after exclusion, the EM OOD ΔECE mean attenuates from -0.0040 to $+0.0015$ (InceptionTime: -0.0198 to -0.0073), indicating the negative EM finding is largely driven by ill-conditioned cells.

Method	Total	Full/6	$\Delta\text{ECE}_{\text{OOD}}$ mean	Neg. dirs
TS	24/30	1/6	$+0.0098$	0
Platt	19/30	1/6	$+0.0088$	1
Isotonic	12/30	0/6	$+0.0009$	3
Vector	17/30	0/6	$+0.0084$	1
Matrix	11/30	0/6	$+0.0041$	3
Dirichlet	16/30	0/6	$+0.0067$	2
EM prior	11/30	0/6	-0.0198	2
BBSE prior	23/30	3/6	$+0.0399$	1

Table 4: Architecture robustness for TS: InceptionTime vs. ResNet1D across 6 transfer pairs, full 5-seed coverage on both architectures. Architecture robustness is supported on both InceptionTime (24/30, 80.0%) and ResNet1D (21/30, 70.0%) with complete 5-seed coverage on all 6 directions. ResNet1D shows direction-specific deficits on CPSC→Chapman and CPSC→PTB-XL (3/5 each) and PTB-XL→CPSC (2/5), not a global architecture deficit.

Transfer pair	InceptionTime	ResNet1D	Total	Note
Chap→CPSC	4/5	4/5	8/10	2 counter-examples
Chap→PTB-XL	4/5	4/5	8/10	2 counter-examples
CPSC→Chap	5/5	3/5	8/10	2 counter-examples
CPSC→PTB-XL	4/5	3/5	7/10	3 counter-examples
PTB-XL→Chap	4/5	5/5	9/10	1 counter-example
PTB-XL→CPSC	3/5	2/5	5/10	5 counter-examples
Total	**24/30**	**21/30**	**45/60**	**75.0%**
Mean ΔECE	+0.0098	+0.0089	+0.0093	
Std ΔECE	0.0108	0.0124	0.0116	ResNet1D/Incep = 1.14
Mean OOD ECE	IT 0.206 → 0.196; RN 0.205 → 0.196		pooled 0.205 → 0.196	raw → TS

Table 5: Predictability LOO R^2 (multi-architecture merged, $n = 6{,}084$ method–shift cells from the post-retrain L2 shift grid; ridge α=1). All methods trigger the failure branch (CI upper bound < 0.5). The three-step chain is downgraded to a two-step chain per the pre-registered failure branch.

Method	R^2	CI lower	CI upper	Verdict
TS	$+0.051$	-0.130	$+0.092$	Failure branch
Platt	-0.008	-0.147	$+0.033$	Failure branch
Isotonic	$+0.010$	-0.108	$+0.046$	Failure branch
Vector	$+0.007$	-0.114	$+0.043$	Failure branch
Matrix	$+0.004$	-0.259	$+0.077$	Failure branch
Dirichlet	$+0.012$	-0.155	$+0.055$	Failure branch
EM prior	$+0.215$	-0.120	$+0.350$	Failure branch
BBSE prior	$+0.254$	-0.003	$+0.378$	Failure branch

Table 6: Deployment criterion sensitivity / specificity on the delivered 780-cell L2 grid (both architectures; beneficial = $\Delta\text{ECE}_{\text{OOD}} < -0.01$ in the post − raw sign convention, opposite to the primary endpoint's raw − TS; P=368, N=412, matching `deployment_metrics.csv`). Threshold search over the same grid (in-sample) and leave-shift-type-out validation; single audit-time input (raw ECE > threshold → predict "deploy TS"). Generated by `scripts/recompute_deployment_780grid.py`.

Scheme	Threshold	Sensitivity	Specificity	J
in-sample	0.206	0.772	0.534	+0.306
leave-out downsample (fs125)	0.209	0.667	0.700	+0.367
leave-out lead drop (leads2)	0.206	0.960	0.486	+0.446
leave-out noise (noise0)	0.206	0.778	0.606	+0.384
leave-out gain (gain2.0)	0.206	0.871	0.414	+0.285
frozen pool-era (0.127)	0.127	0.929	0.228	+0.158

Table 7: Trivial strategy comparison. Mean ΔECE; negative = beneficial (ECE reduction). always-none and always-TS span all 780 (method, shift) evaluations; lookup-table routes a row to TS iff its audit-time raw ECE exceeds the 780-grid in-sample-optimal threshold 0.206 (476/780 = 61% of rows routed; `scripts/recompute_deployment_780grid.py` on `results/l2_shift_full_780cells_detail.csv`). The always-buy-n-labels row is a historical sizing reference from the archived pool-era reference grid, not part of the delivered 780-cell comparison on which the other rows are computed.

Strategy	Mean ΔECE	Std	n
always-none	0.0000	0.0000	780
always-TS	−0.0089	0.0116	780
always-buy-n-labels	−0.25	0.15	1230
lookup-table	−0.0116	0.0116	476

Table 8: Small-sample recalibration dose-response (protocol §5 S2): mean SmoothECE benefit (ECE reduction) over the 60 transfer-grid cells $\times$ 20 patient-level repeats, evaluated on the held-out complement; the fraction of cells with positive mean benefit as a raised percentage (*superscripts read as percent, not exponents*). ts_s1 is the zero-shot source-fitted temperature control (uses no budget; rows differ only through the shrinking complement). Isotonic at $n_{\text{cal}} = 50$ covers $40/60$ cells (pre-registered small-sample guard $n_{\text{cal}} \leq 10K$).

n_{cal}	50	100	250	500	1000
ts_s1 (0-shot)	$+0.0093^{75\%}$	$+0.0093^{75\%}$	$+0.0093^{75\%}$	$+0.0093^{75\%}$	$+0.0093^{75\%}$
ts (refit)	$+0.1269^{97\%}$	$+0.1330^{98\%}$	$+0.1382^{98\%}$	$+0.1390^{98\%}$	$+0.1394^{98\%}$
platt	$+0.1502^{100\%}$	$+0.1576^{100\%}$	$+0.1606^{100\%}$	$+0.1626^{100\%}$	$+0.1607^{100\%}$
dirichlet	$+0.0629^{75\%}$	$+0.1313^{100\%}$	$+0.1647^{100\%}$	$+0.1720^{100\%}$	$+0.1736^{100\%}$
isotonic	$+0.1352^{97\%}$	$+0.1468^{100\%}$	$+0.1617^{100\%}$	$+0.1650^{100\%}$	$+0.1644^{100\%}$

Table 9: The fifteen counter-examples, cell by cell. ID order is the canonical order of `results/counterexample_discrimination.json` (the artifact the manuscript cites); per-cell narratives for the twelve non-tabulated paragraphs ship in the Supplementary Materials. ΔECE and 95% CIs from the primary-endpoint table; OOD accuracy from the per-experiment checkpoint artifacts; T_{src} = source-fitted temperature. Class census: seven "low acc" (OOD accuracy ≤ 0.50), eight seed-level temperature mis-transfers, of which CE6 is the over-correction case.

ID	Cell	$\Delta\text{ECE}_{\text{OOD}}$	95% CI	OOD acc	T_{src}	Class
CE1	Chap→CPSC / IT / s45	-0.0056	$[-0.0058, -0.0055]$	0.4711	0.970	low acc
CE2	Chap→CPSC / RN / s43	-0.0006	$[-0.0006, -0.0006]$	0.4779	0.997	low acc
CE3	Chap→PTB-XL / IT / s42	-0.0025	$[-0.0035, -0.0016]$	0.4655	0.984	low acc
CE4	Chap→PTB-XL / RN / s45	-0.0036	$[-0.0036, -0.0035]$	0.5041	0.988	T mis-transfer
CE5	CPSC→Chap / RN / s42	-0.0010	$[-0.0012, -0.0008]$	0.6147	0.993	T mis-transfer
CE6	CPSC→Chap / RN / s44	-0.0168	$[-0.0170, -0.0165]$	0.5725	0.932	over-correction
CE7	CPSC→PTB-XL / IT / s43	-0.0057	$[-0.0058, -0.0054]$	0.4656	0.975	low acc
CE8	CPSC→PTB-XL / RN / s43	-0.0050	$[-0.0051, -0.0049]$	0.3877	0.980	low acc
CE9	CPSC→PTB-XL / RN / s44	-0.0049	$[-0.0049, -0.0048]$	0.3929	0.981	low acc
CE10	PTB-XL→Chap / IT / s44	-0.0033	$[-0.0034, -0.0032]$	0.4956	0.990	low acc
CE11	PTB-XL→CPSC / IT / s42	-0.0083	$[-0.0089, -0.0074]$	0.5839	0.967	T mis-transfer
CE12	PTB-XL→CPSC / IT / s45	-0.0123	$[-0.0132, -0.0106]$	0.6101	0.953	T mis-transfer
CE13	PTB-XL→CPSC / RN / s42	-0.0069	$[-0.0079, -0.0052]$	0.5615	0.970	T mis-transfer
CE14	PTB-XL→CPSC / RN / s45	-0.0057	$[-0.0059, -0.0054]$	0.5746	0.978	T mis-transfer
CE15	PTB-XL→CPSC / RN / s46	-0.0048	$[-0.0051, -0.0041]$	0.5727	0.981	T mis-transfer

Table B.1: Artifact index: repository artifacts ↔ manuscript sections. All paths relative to the repository root.

Artifact	Used in
`results/robustness_validation_5seeds.csv` (primary-endpoint table, 60 cells)	Sec. 5.1, Table 1, Table 4
`results/samplelevel_ts_60cells.csv` (adaptive-TS per-cell deliverable)	Sec. 5.3 (ATS paragraph)
`results/c4_resnet1d_8method.csv` (8-method delivered rows)	Sec. 5.3, Table 3
`results/robustness_val_8method_perbranch_recheck.csv` (per-branch correctness recheck)	Sec. 5.3, semantics notice
`results/deployment_780grid_recompute.csv` (+ 780-cell detail CSV)	Sec. 5.6, Tables 6, 7
`results/s2_dose_response.csv` (labeling-budget dose–response)	Sec. 5.7, Table 8
`results/noise_floor_curve.csv`	Sec. 5.8
`results/counterexample_discrimination.csv`, `.json`	Sec. 6 (discrimination experiment)
`results/predictability_corrected_v2.csv`, `results/predictability_3arch.csv`	Sec. 5.5, Table 5
`results/c1_dcr_ncv_multiclass.csv`, `results/c1_dcr_ncv_summary.csv`	Sec. 5.9
`results/e3_patient_bca_cells.csv`, `results/e3_patient_bca_summary.json`	Sec. 5.9 (registered patient-level unit)
`results/real_prior_recovery.csv/_summary.json`	Honesty item 1
`results/discrimination_metrics_60exp.csv` (AUROC/AUPRC)	Sec. 6 (discrimination block)
`results/decomposition_validation_n20000.csv`	Sec. 5.4
`results/rebuild_shapley_{crossfit,slopeT}.{csv,json}`	Sec. 5.1
`scripts/train.py`, `scripts/export_splits.py` (leak-fixed code line)	Sec. 4.1, post-registration repair
`scripts/verify_release_results.py` (12-check release gate)	repository; cited in Code availability
`scripts/recheck_method_semantics.py` (+ samplelevel, deployment recompute scripts)	Sec. 5.3, Sec. 5.6
`docs/EXPERIMENT_PROTOCOL.md`, amendments A1–A4	Sec. 4, Appendix A